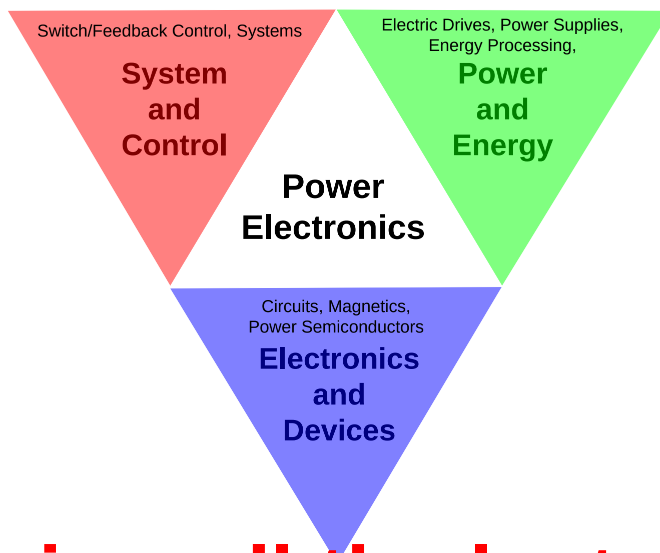
Power Electronics

Week 5 and week 6 lectures review



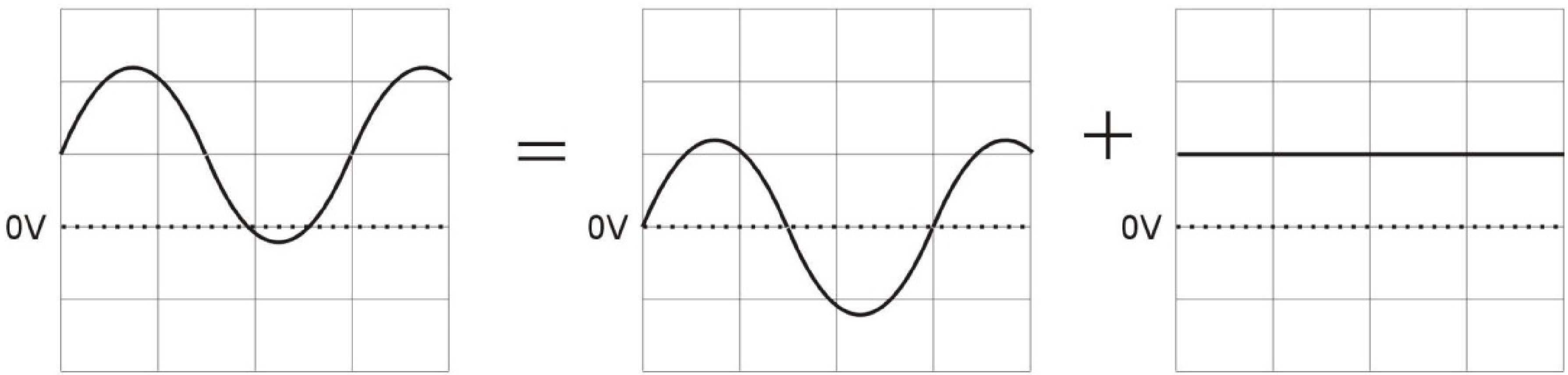
Please review all the lecture materials, from Lecture 1 to Lecture 6

Question:

Show that the Root Mean Square (RMS) value of a sinusoidal waveform with peak amplitude A volts and DC offset B volts is given by:

$$V_{RMS} = \sqrt{\frac{A^2}{2} + B^2}$$

$$V_{TOTAL\,rms} = \sqrt{\left(V_{ACrms}{}^2 + V_{DCrms}{}^2\right)}$$



Sinewave + DC Offset | Sinewave | DC Offset

$$A\sin(\omega t) + B \qquad A\sin(\omega t) \qquad B$$

Question:

Show that the RMS value of a sinusoidal waveform with peak amplitude A volts and DC offset B volts is given by:

$$V_{RMS} = \sqrt{\frac{A^2}{2} + B^2}$$

Solution:

$$Let\ \mathrm{v(t) = Asin(}\omega\mathrm{t) + B. Then}$$

$$\mathrm{v(t)}^2 = A^2 \sin^2(\omega t) + 2AB\sin(\omega t) + B^2$$

$$V_{RMS}^2 = \frac{1}{2\pi}\int_0^{2\pi} v(t)^2\,dt = \frac{1}{2\pi}\int_0^{2\pi}\left(A^2\sin^2(\omega t) + 2AB\sin(\omega t) + B^2\right)d\omega t$$

$$= \frac{1}{2\pi}\int_0^{2\pi}\left(\frac{A^2}{2}\left(1-\cos(2\omega t)\right) + 2AB\sin(\omega t) + B^2\right)d\omega t$$

$$= \frac{1}{2\pi}\left[\left[\frac{A^2\omega t}{2}\right]_0^{2\pi} - \left[\frac{A^2\omega t}{4}\sin(2\omega t)\right]_0^{2\pi} - \left[2AB\cos(\omega t)\right]_0^{2\pi} + \left[B^2\omega t\right]_0^{2\pi}\right]$$

$$= \frac{1}{2\pi}\left[\frac{2\pi\,A^2}{2} + 2\pi\,B^2 - 2AB + 2AB\right] = \frac{\mathrm{A}^2}{2} + B^2$$

$$=> V_{RMS} = \sqrt{\frac{\mathrm{A}^2}{2} + B^2}$$

Form Factor （FF）波形因数

The ratio of the Root Mean Square (RMS) to average value of a periodic function is called the Form Factor, F, of the function.

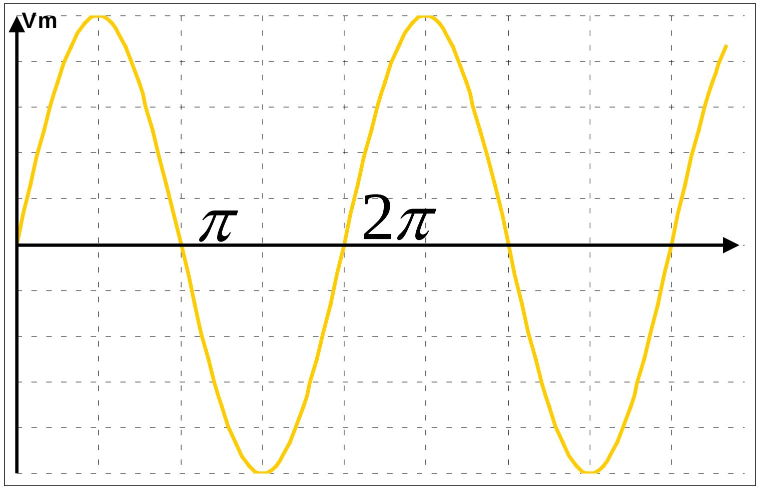
$$\textbf{Form Factor} = \frac{Y_{rms}}{Y_{ave}} = \frac{\sqrt{\frac{1}{T}\int_0^T |y(t)|^2\, dt}}{\frac{1}{T}\int_0^T |y(t)|\, dt}$$

Don’t forget the abs

- The form factor is important because the **heating effect** (and hence the temperature rise) in circuit components, transformer windings, etc. is proportional to the square of the RMS current. Measurement instruments also frequently assume a sinusoidal signal, and high form factor waveforms can lead to significant measurement errors.
- It identifies the ratio of the direct current of equal power relative to the given alternating current

Example: Calculate the form factor for a sinusoidal voltage waveform.

Let the voltage be $v = V_m \sin(\omega t)$



$$v_{ave} = \frac{1}{\pi}\int_0^{\pi} V_m \sin(\omega t) d(\omega t) = \frac{V_m}{\pi}\int_0^{\pi} \sin(\omega t) d(\omega t)$$

$$= \frac{V_m}{\pi}\left[-\cos(\omega t)\right]_0^{\pi} = -\frac{V_m}{\pi}\cos(\pi) + \frac{V_m}{\pi}\cos(0) = \frac{2V_m}{\pi}$$

$$= 0.64V_m$$

$$v_{rms} = \sqrt{\frac{1}{\pi}\int_0^{\pi}\left[V_m \sin(\omega t)\right]^2 d(\omega t)} = \sqrt{\frac{V_m^{\ 2}}{\pi}\int_0^{\pi}\left[\frac{1}{2}\left(1-\cos(2\omega t)\right)\right] d(\omega t)}$$

$$= \sqrt{\frac{V_m^{\ 2}}{2\pi}\left[\omega t\right]_0^{\pi} - \frac{V_m^{\ 2}}{2\pi}\left[\frac{1}{2}\sin(2\omega t)\right]_0^{\pi}} = \sqrt{\frac{V_m^{\ 2}}{2}} = \frac{V_m}{\sqrt{2}} = 0.71V_m$$

$$F = \frac{v_{rms}}{v_{ave}} = \frac{\frac{V_m}{\sqrt{2}}}{\frac{2V_m}{\pi}} = \frac{\frac{\pi V_m}{\sqrt{2}}}{2V_m} = \frac{\pi V_m}{2\sqrt{2}V_m} = \frac{\pi}{2\sqrt{2}} = 1.111$$

If you ever get a form factor less than one your maths is incorrect.

The RMS of the voltage which is the sum of more than two periodic voltages

If a periodic voltage is the sum of two periodic voltage waveforms, $v(t) = v_1(t) + v_2(t)$, the rms value of $v(t)$ is determined from Eq. (2-37) as

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T (v_1 + v_2)^2\, dt = \frac{1}{T}\int_0^T \left(v_1^2 + 2v_1 v_2 + v_2^2\right) dt$$

or

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T v_1^2\, dt + \frac{1}{T}\int_0^T 2v_1 v_2\, dt + \frac{1}{T}\int_0^T v_2^2\, dt$$

The term containing the product $v_1 v_2$ in the above equation is zero if the functions v_1 and v_2 are orthogonal. A condition that satisfies that requirement occurs when v_1 and v_2 are sinusoids of different frequencies. For orthogonal functions,

$$V_{\text{rms}}^2 = \frac{1}{T}\int_0^T v_1^2(t)\, dt + \frac{1}{T}\int_0^T v_2^2(t)\, dt$$

Noting that

$$\frac{1}{T}\int_0^T v_1^2(t)\, dt = V_{1,\text{rms}}^2 \quad \text{and} \quad \frac{1}{T}\int_0^T v_2^2(t)\, dt = V_{2,\text{rms}}^2$$

then

$$V_{\text{rms}} = \sqrt{V_{1,\text{rms}}^2 + V_{2,\text{rms}}^2}$$

If a voltage is the sum of more than two periodic voltages, all orthogonal, the rms value is

$$\boxed{V_{\text{rms}} = \sqrt{V_{1,\text{rms}}^2 + V_{2,\text{rms}}^2 + V_{3,\text{rms}}^2 + \ \ldots} = \sqrt{\sum_{n=1}^{N} V_{n,\text{rms}}^2}}$$

Similarly,

$$\boxed{I_{\text{rms}} = \sqrt{I_{1,\text{rms}}^2 + I_{2,\text{rms}}^2 + I_{3,\text{rms}}^2 + \ \ldots} = \sqrt{\sum_{n=1}^{N} I_{n,\text{rms}}^2}}$$

Determine the effective (rms) value of $v(t) = 4 + 8\sin(\omega_1 t + 10°) + 5\sin(\omega_2 t + 50°)$ for (*a*) $\omega_2 = 2\omega_1$ and (*b*) $\omega_2 = \omega_1$.

■ Solution

(*a*) The rms value of a single sinusoid is $V_m/\sqrt{2}$, and the rms value of a constant is the constant. When the sinusoids are of different frequencies, the terms are orthogonal and Eq. (2-39) applies.

$$V_{\text{rms}} = \sqrt{V_{1,\text{rms}}^2 + V_{2,\text{rms}}^2 + V_{3,\text{rms}}^2} = \sqrt{4^2 + \left(\frac{8}{\sqrt{2}}\right)^2 + \left(\frac{5}{\sqrt{2}}\right)^2} = 7.78 \text{ V}$$

(*b*) For sinusoids of the same frequency, Eq. (2-39) does not apply because the integral of the cross product over one period is not zero. First combine the sinusoids using phasor addition:

$$8\angle 10° + 5\angle 50° = 12.3\angle 25.2°$$

The voltage function is then expressed as

$$v(t) = 4 + 12.3 \sin(\omega_1 t + 25.2°) \text{ V}$$

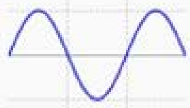
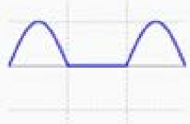
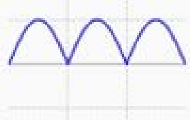
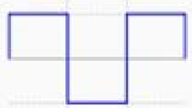
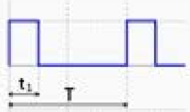
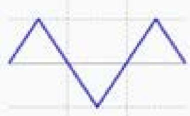
The rms value of this voltage is determined from Eq. (2-39) as

$$V_{\text{rms}} = \sqrt{4^2 + \left(\frac{12.3}{\sqrt{2}}\right)^2} = 9.57 \text{ V}$$

Typical Form Factor

Different waveforms have different ratio between its RMS value and its average value – ***Form Factor***

DC:
form factor=1

Waveform	Image	RMS	ARV	Form Factor
Sine wave		$\frac{a}{\sqrt{2}}$[2]	$a\frac{2}{\pi}$[2]	$\frac{\pi}{2\sqrt{2}} \approx 1.11072073$[3]
Half-wave rectified sine		$\frac{a}{\sqrt{2}}$	$\frac{a}{\pi}$	$\frac{\pi}{\sqrt{2}} \approx 2.22144146$
Full-wave rectified sine		$\frac{a}{\sqrt{2}}$	$a\frac{2}{\pi}$	$\frac{\pi}{2\sqrt{2}}$
Square wave, constant value		a	a	$\frac{a}{a} = 1$
Pulse wave		$a\sqrt{D}$[7]	aD	$\frac{1}{\sqrt{D}} = \sqrt{\frac{T}{\tau}}$
Triangle wave		$\frac{a}{\sqrt{3}}$[8]	$\frac{a}{2}$	$\frac{2}{\sqrt{3}} \approx 1.15470054$

Examples:

Assuming a voltage signal is given as $V_1 = 10\sin(100\pi t) + 10$

1) Calculate the RMS value of the voltage signal. [3]
2) Calculate the form factor for the voltage signal. [2]

Answers:

1) RMS: $I_{rms} = \sqrt{\frac{10^2}{2} + 10^2} = 12.25$

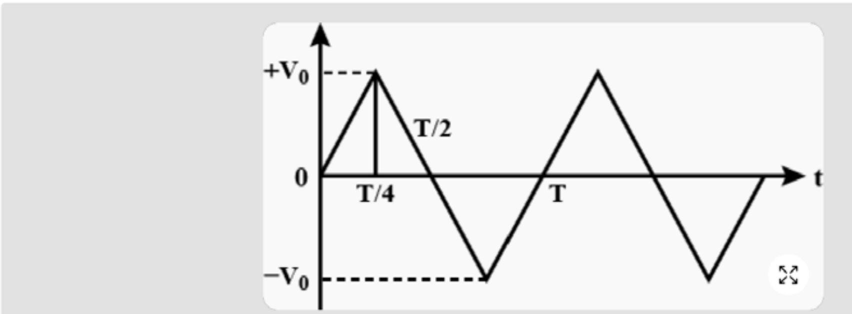
2)

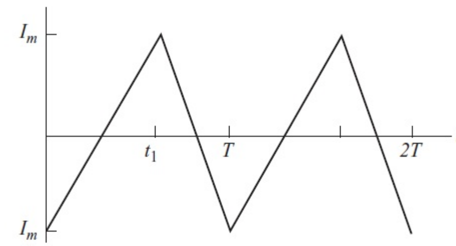
$$I_{|avg|} = \frac{1}{2\pi}\int_0^{2\pi}\left(\left|10\sin(100\pi t) + 10\right|\right)dt = 10$$

FF=1.225

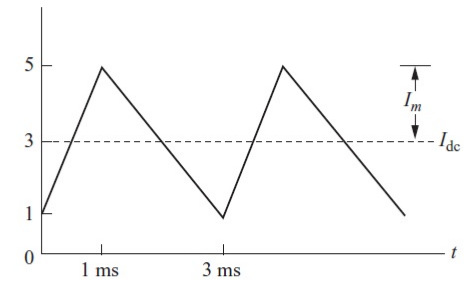
Examples

The voltage time $(V - t)$ graph for triangular wave having peak value V_0 is as shown in figure. The rms value of V in time interval from $t = 0$ to $T/4$ is:





(a)



(b)

Answer

Correct option is $\frac{V_0}{\sqrt{3}}$

The peak value of the triangular wave is V_0.

The equation of the line in the interval 0 to $\frac{T}{4}$ is given as,

$$V = \frac{4V_0}{T}t$$

The rms value of the triangular wave is given as,

$$V_{rms} = \sqrt{\frac{4}{T}\int_0^{\frac{T}{4}} (V)^2 dt}$$

$$= \sqrt{\frac{4}{T} \times \left(\frac{4V_0}{T}\right)^2 \int_0^{\frac{T}{4}} (t)^2 dt}$$

$$= \sqrt{\left(\frac{64V_0^2}{T^3}\right)\left(\frac{t^3}{3}\right)_0^{\frac{T}{4}}}$$

$$= \frac{V_0}{\sqrt{3}}$$

Thus the rms value of V in the time interval 0 to $\frac{T}{4}$ is $\frac{V_0}{\sqrt{3}}$.

$$V_{TOTAL\,rms} = \sqrt{\left(V_{ACrms}{}^2 + V_{DCrms}{}^2\right)}$$

(*a*) The details of the integration are quite long, but the result is simple: The rms value of a triangular current waveform is

$$I_{rms} = \frac{I_m}{\sqrt{3}}$$

(*b*) The rms value of the offset triangular waveform can be determined by using the result of part (*a*). Since the triangular waveform of part (*a*) contains no dc component, the dc signal and the triangular waveform are orthogonal, and Eq. (2-40) applies.

$$I_{rms} = \sqrt{I_{1,rms}^2 + I_{2,rms}^2} = \sqrt{\left(\frac{I_m}{\sqrt{3}}\right)^2 + I_{dc}^2} = \sqrt{\left(\frac{2}{\sqrt{3}}\right)^2 + 3^2} = 3.22 \text{ A}$$

Power and Energy in Resistors, Inductors and Capacitors.

For a resistor R, power $p = iv = i^2R = v^2/R$.
A resistor cannot store energy.

- Oppose any change in current
- Whereas resistors oppose current flow.
- Inductor store energy as magnetic field

For an inductor L, power $p = iv = iL\frac{di}{dt}$

And the energy stored in the inductor is $\boxed{W = \int p\,dt = \int iL\,di = \frac{1}{2}Li^2}$

DANGER!!!

Charged inductor ($i \neq 0$) do not allowed to be **open**. $di/dt \rightarrow \infty$, $v \rightarrow \infty$ will damage circuit components, leads to overhigh discharging power

- Stores energy in electric field
- Resist the voltage change
- Keeps the charge if open-circuited

For a capacitor C, power $p = iv = vC\frac{dv}{dt}$

And the energy stored in the capacitor is $\boxed{W = \int p\,dt = \int Cv\,dv = \frac{1}{2}Cv^2}$

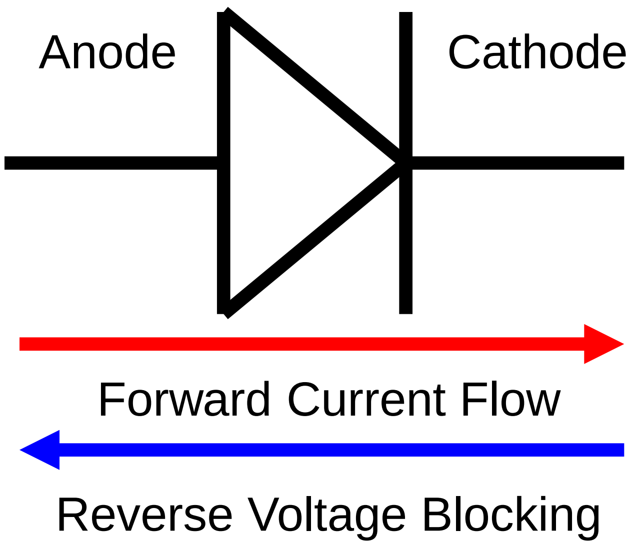
DANGER!!!

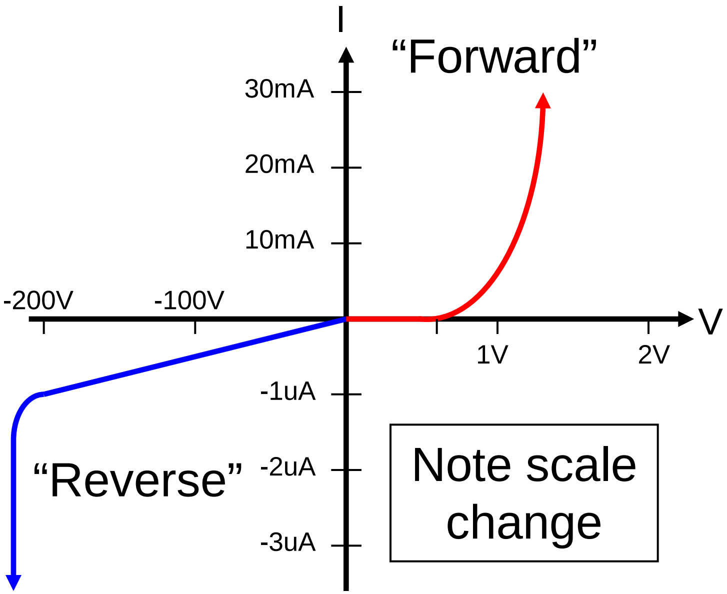
Charged capacitors ($v \neq 0$) do not allowed to be **short**. $dv/dt \rightarrow \infty$, $i \rightarrow \infty$ will damage the capacitor, leads to overhigh discharging power p.

In the case of the resistor there is an energy conversion process in that the power supplied is converted into heat. For the inductor and capacitor there is no conversion: the energy supplied is stored (in a magnetic or electric field respectively), and is available for return to the circuit if the conditions change.

Diodes

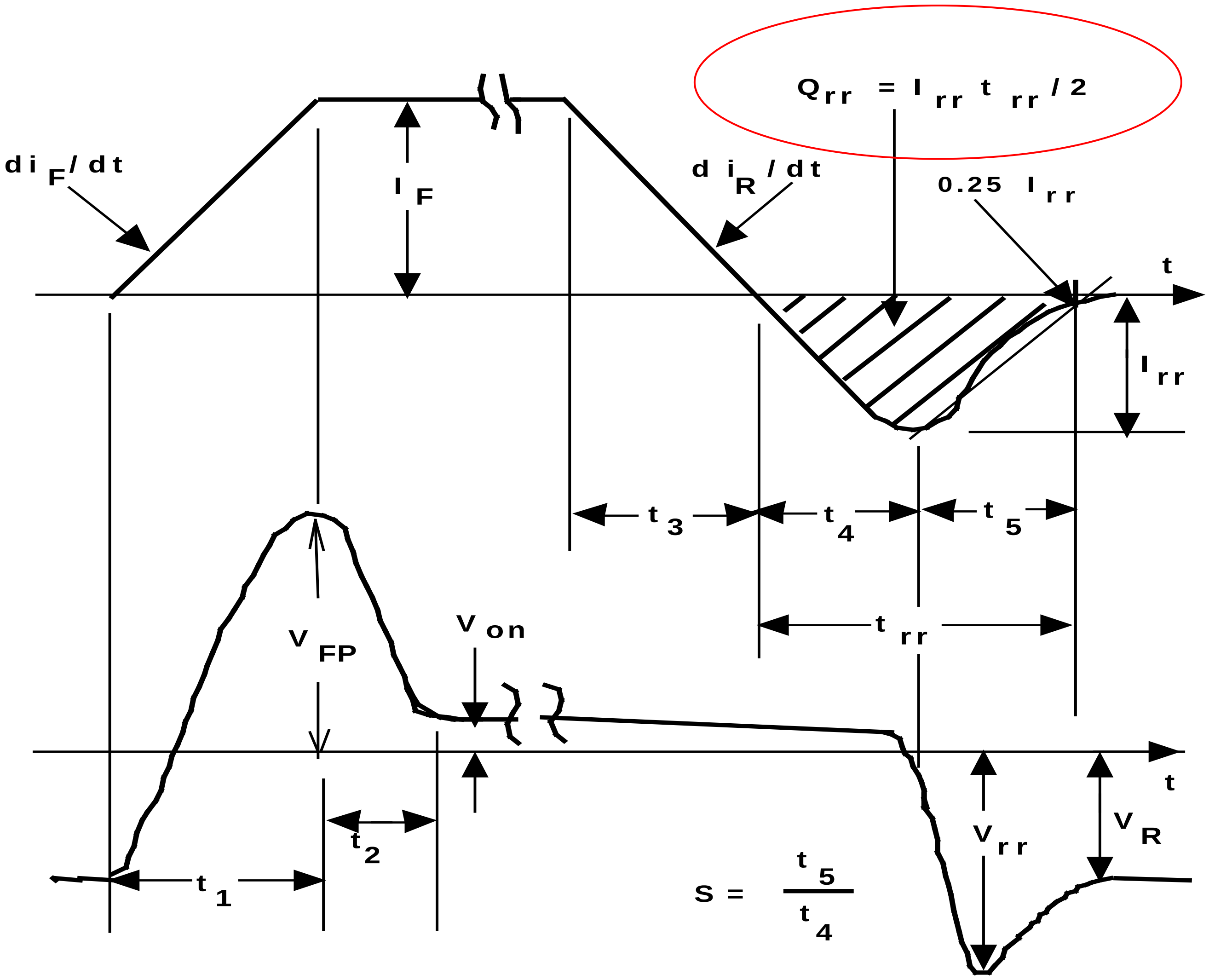
All the circuit elements we have considered so far have been linear – doubling the applied signal (voltage, say) doubles the response (a current, say). The diode is a non-linear semiconductor device with the V-I curve as shown in the diagram.





The diode's arrow indicates the normal current flow. In operation, the diode behaves like a voltage-sensitive switch. When the anode is more positive than the cathode current flows (quadrant 1 of diagram). When the anode is negative with respect to the cathode, current flow is blocked (quadrant 3 of diagram). This selective process of passage / blocking of current is called *rectification*.

Reverse Recovery Charge Q_{rr}



1. I_F(Forward current)
2. V_{FP}(peak forward voltage)
3. V_{on}(forward voltage drop)
4. V_{rr}(maximum reverse voltage)
5. V_R(reverse voltage)
6. I_{rr}(maximum reverse current)
7. (t_1+t_2) (turn on transient time)
8. (t_3+t_4+t_5) (turn off transient time)
9. (t_4+t_5) (reverse recovery time)
10. $Q_{rr}=I_{rr}*t_{rr}/2$
11. $S=t_5/t_4$ (recovery factor)

Example:

A diode in a power supply circuit is operated with a forward current I_F = 10A and a forward voltage drop V_F at this current of 1.1V. The power supply circuit is switching at a frequency of 31.5KHz and the duty cycle is 50%. The steady-state reverse voltage across the diode is -50V and the reverse recovery charge Q_{rr} is 2.5 µC.

What power does the diode dissipate?

$$P_{Total} = P_{Conduction} + P_{Switching}$$

Conduction loss = V_F x I_F x 0.5 = 1.1V x 10A x 50% = 5.5W.

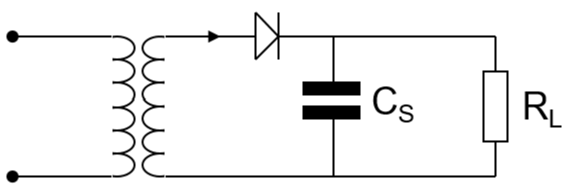
Switching loss = Q_{rr} x V_R x F_{SW} = $2.5\text{x}10^{-6}$C x 50V x 31500 = 3.9W

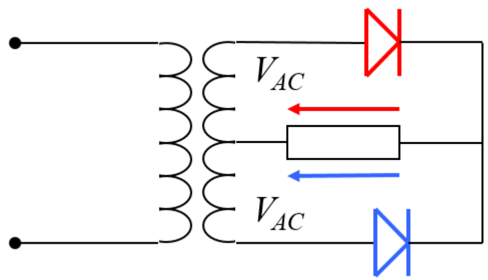
$\Rightarrow$Total power loss = 5.5W + 3.9W = <u>9.4W</u>

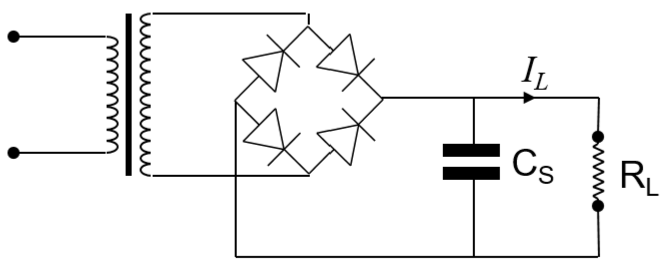
Important:
Switching losses vary directly with switching frequency.
Conduction losses vary directly with 'on' time (and current, obviously).

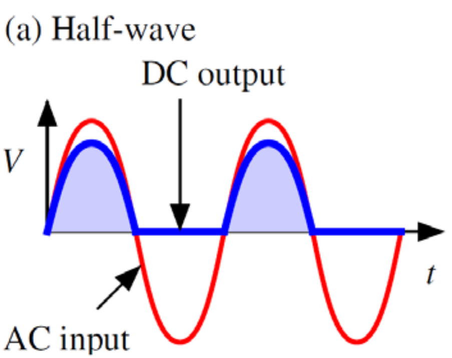
Half-wave, Centre-tapped full-wave, Bridge full-wave rectifiers

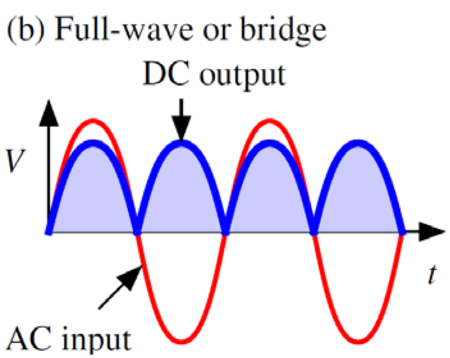
	half-wave	full-wave	bridge
windings needed	one	two (centre-tapped)	one
utlilization	half	half (one winding at a time)	full
diodes needed	1	2	4
reverse rating	V_{peak}	$2V_{\text{peak}}$	V_{peak}
voltage drop	0.7 V	0.7 V	1.4 V
frequency of ripple	f_{input}	$2f_{\text{input}}$	$2f_{\text{input}}$
ease of smoothing	difficult (empty half cycles)	easier	easier

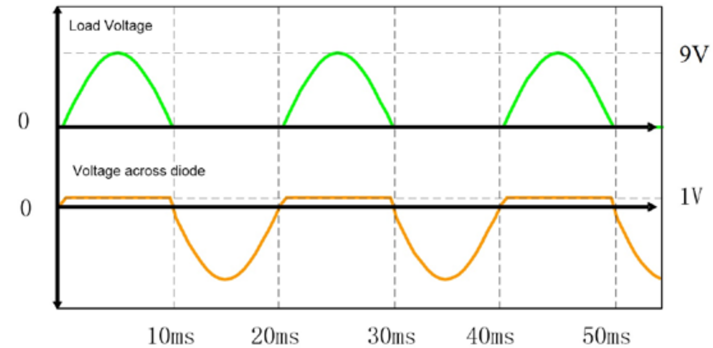






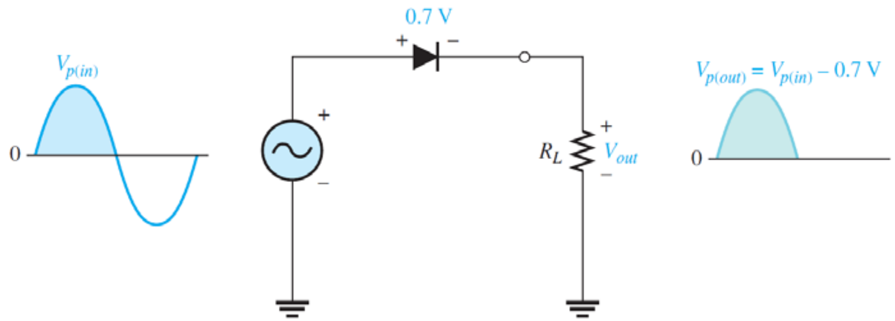


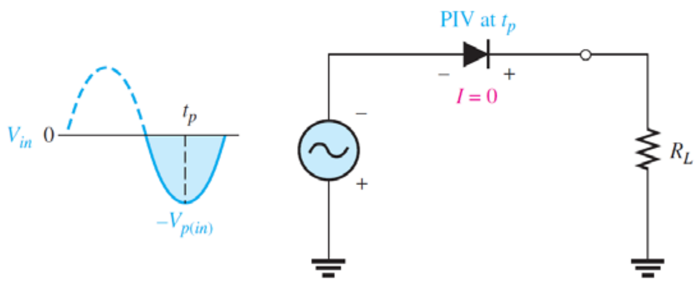




Half-Wave Rectifier

$$V_{p(out)} = V_{P(in)} - 0.7\text{ V}$$

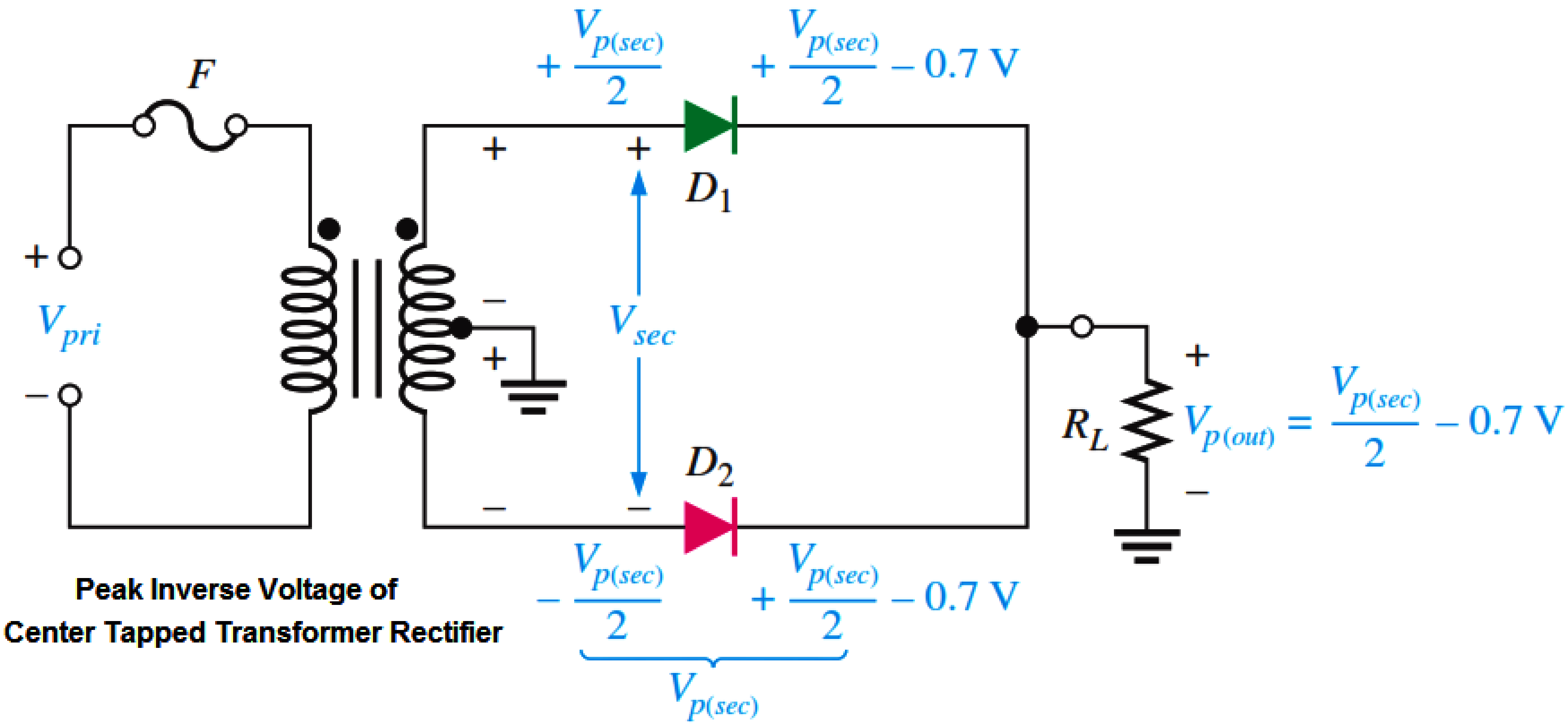




PIV=V$_{p(in)}$

Peak Inverse Voltage (PIV)

Centre-Tapped Full-wave Rectifier



Peak Inverse Voltage of Center Tapped Transformer Rectifier

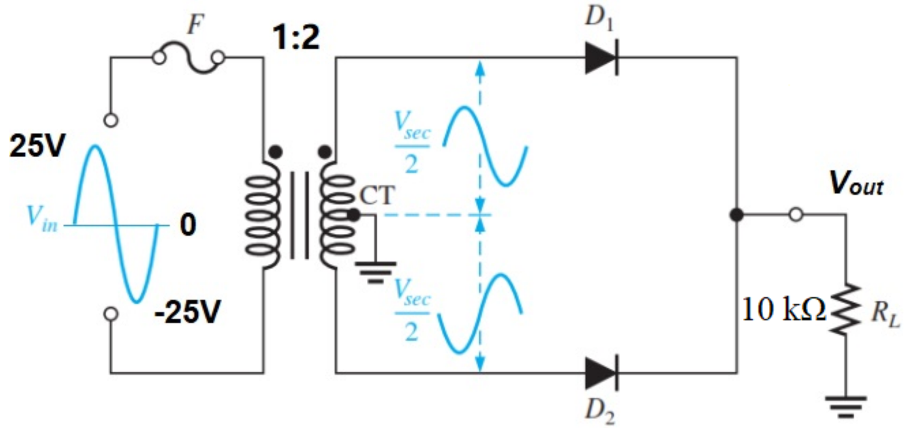
$$\begin{aligned}\text{PIV} &= \left(\frac{V_{p(sec)}}{2} - 0.7\ \text{V}\right) - \left(-\frac{V_{p(sec)}}{2}\right) = \frac{V_{p(sec)}}{2} + \frac{V_{p(sec)}}{2} - 0.7\ \text{V}\\ &= V_{p(sec)} - 0.7\ \text{V}\end{aligned}$$

Since $V_{p(out)} = V_{p(sec)}/2 - 0.7\ \text{V},$

$$V_{p(sec)} = 2V_{p(out)} + 1.4\ \text{V}$$

$$\mathbf{PIV = 2V_{p(out)} + 0.7\ V}$$

$$I_{p(L)} = \frac{V_{p(out)}}{R_L} = \frac{V_{p(\text{sec})}/2 - 0.7}{R_L}$$

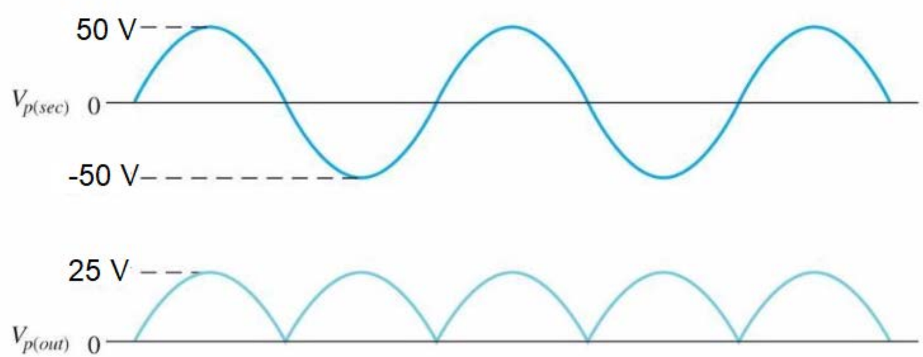


For ideal diodes, a 25 V peak sine wave is applied to the primary winding.

(1) Show the voltage waveforms V_{out} across the load R_L.

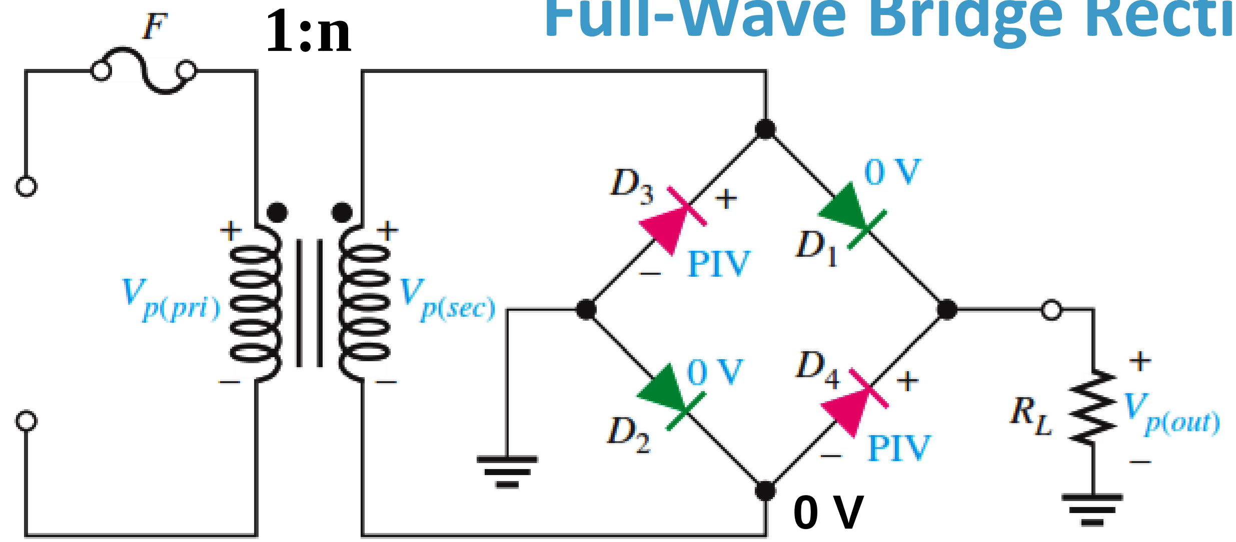
(2) What the minimum PIV rating must the diodes have?

(a)



(b) $$PIV = 2V_{p(out)} = 2(25V)=50\text{ V}$$

Full-Wave Bridge Rectifier

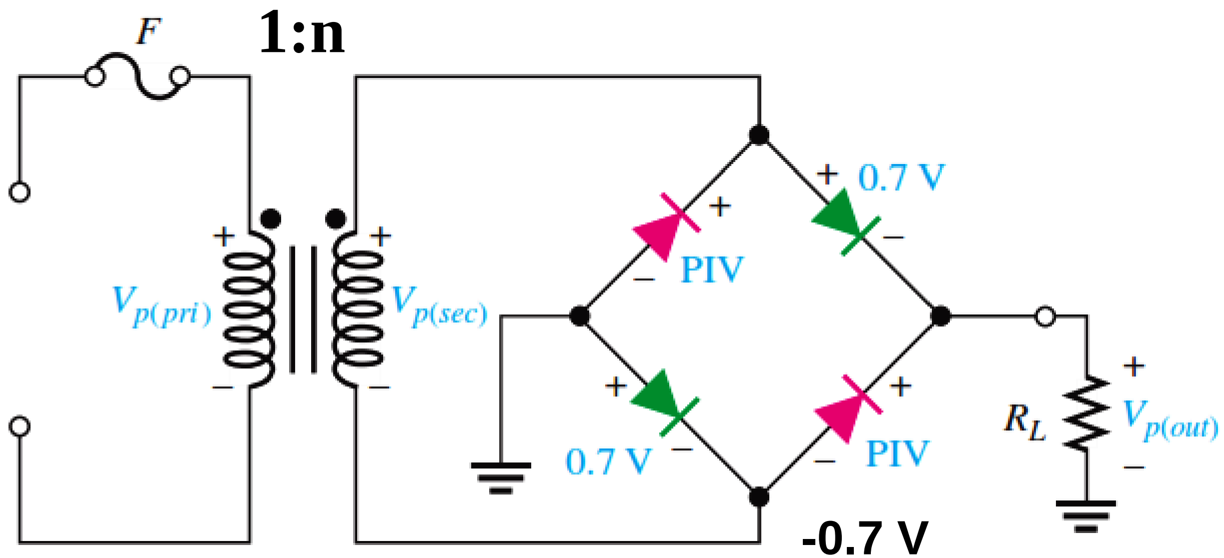


(a) For the ideal diode model (forward-biased diodes D_1 and D_2 are shown in green), PIV = $V_{p(out)}$.

$$V_{sec} = nV_{pri}$$

$$V_{p(out)} = V_{p(sec)}$$

$$\text{PIV} = V_{p(out)}$$



(b) For the practical diode model (forward-biased diodes D_1 and D_2 are shown in green), PIV = $V_{p(out)}$ + 0.7 V.

$$V_{sec} = nV_{pri}$$

$$V_{p(out)} = V_{p(sec)} - 0.7$$

$$\text{PIV} = V_{p(sec)}$$

Ideal Power Diode Full-Wave Bridge Rectifier

Bridge Output Voltage From transformer theory, the secondary voltage is equal to the primary voltage times the turns ratio as stated by the equation:

$$V_{sec} = nV_{pri}$$

This equation can be applied because both V_{sec} and V_{pri} are peak values. Because V_{in} is the same as the primary voltage (as long as the fuse is good), the peak secondary voltage can be expressed as $V_{p(sec)} = nV_{p(in)}$.

As you can see in Figure 35, two diodes are always in series with the load resistor during both the positive and the negative half-cycles. Neglecting the barrier potentials of the two diodes, the output voltage is a full-wave rectified voltage with a peak value equal to the peak secondary voltage.

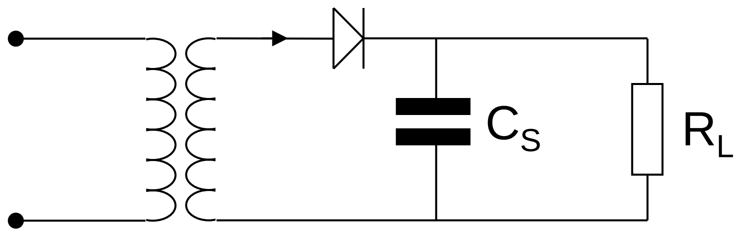
$$\boldsymbol{V_{p(out)} = V_{p(sec)}}$$

Peak Inverse Voltage (PIV) Let's assume that the input is in its positive half-cycle when D_1 and D_2 are forward-biased and examine the reverse voltage across D_3 and D_4. In Figure 36, you can see that D_3 and D_4 have a peak inverse voltage equal to the peak secondary voltage, $V_{p(sec)}$. Since the peak secondary voltage is equal to the peak output voltage, the peak inverse voltage is

$$\mathbf{PIV} = \boldsymbol{V_{p(out)}}$$

The PIV rating of the bridge diodes is half that required for the center-tapped configuration for the same output voltage.

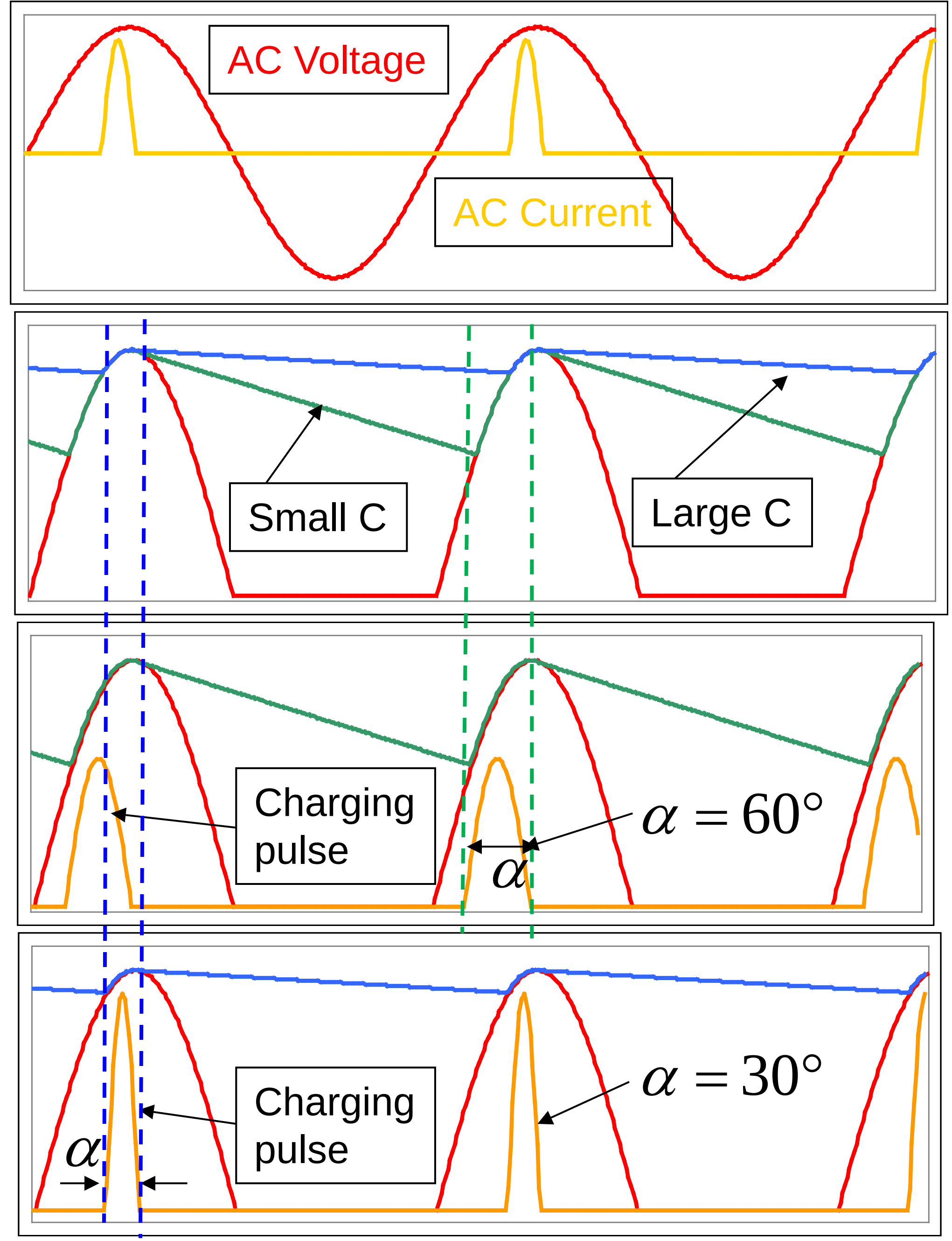
Half-wave rectification with capacitive smoothing



In this circuit the diode only conducts when the anode is more positive than the capacitor voltage.

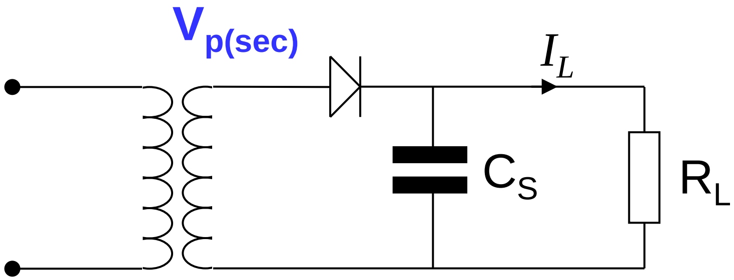
For a given load power, a large capacitor results in a lower ripple voltage. However, this leads to a shorter charging current pulse and hence the peak current through the diode increases. The form factor also becomes higher (typically 1.8 to 2.2) and thus the transformer requires a larger wire diameter and higher VA rating to cope without overheating. The diode conduction angle is α .

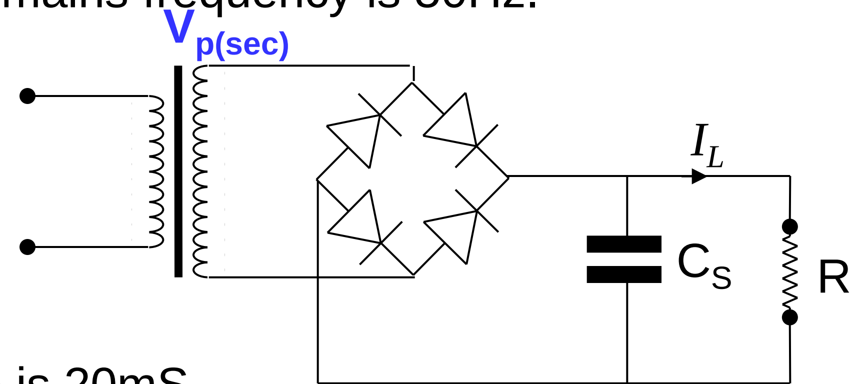
Harmonic currents are high.



Example.

What is the peak to peak ripple voltage in the circuits shown for a load current of 5A and a capacitor of 15,000μF? Assume the mains frequency is 50Hz.

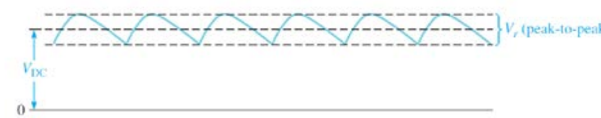




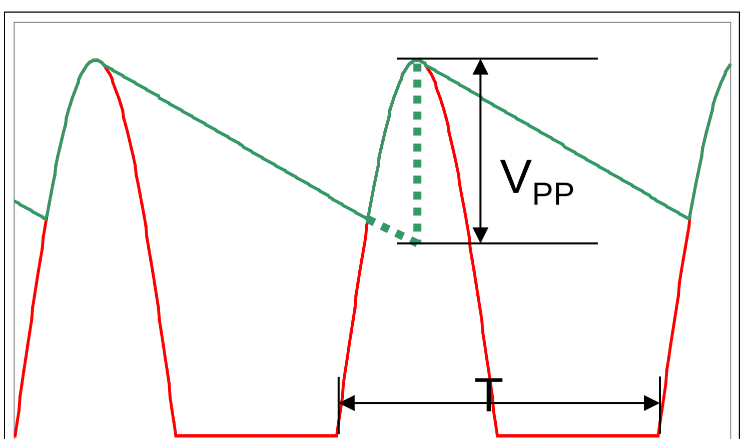
For the half-wave case, the discharge time is 20mS.

PIV=$V_{p(sec)}$= $V_{p(load)}$

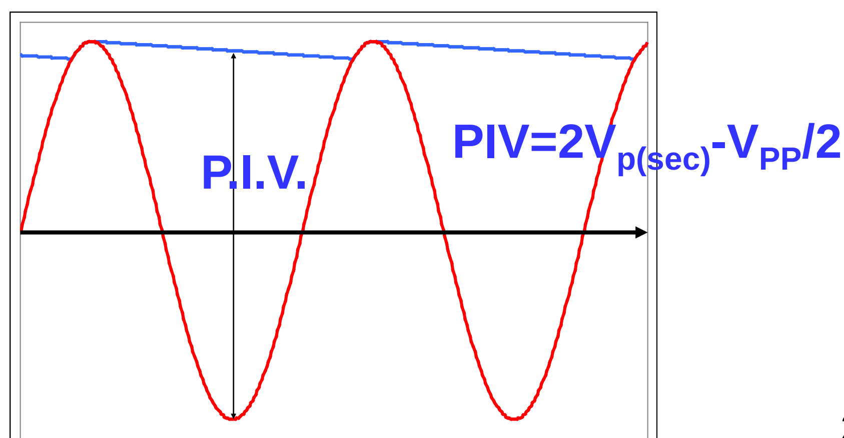
$$\frac{dV_{Load}}{dt} = \frac{I_{Load}}{C} = \frac{5}{15000\times10^{-6}} = 333\ Vs^{-1}$$



$$=> \Delta V_{PP} = \text{Discharge time} \times \frac{dV_L}{dt} = 20mS \times 333Vs^{-1} = 6.67V$$



Half-wave case

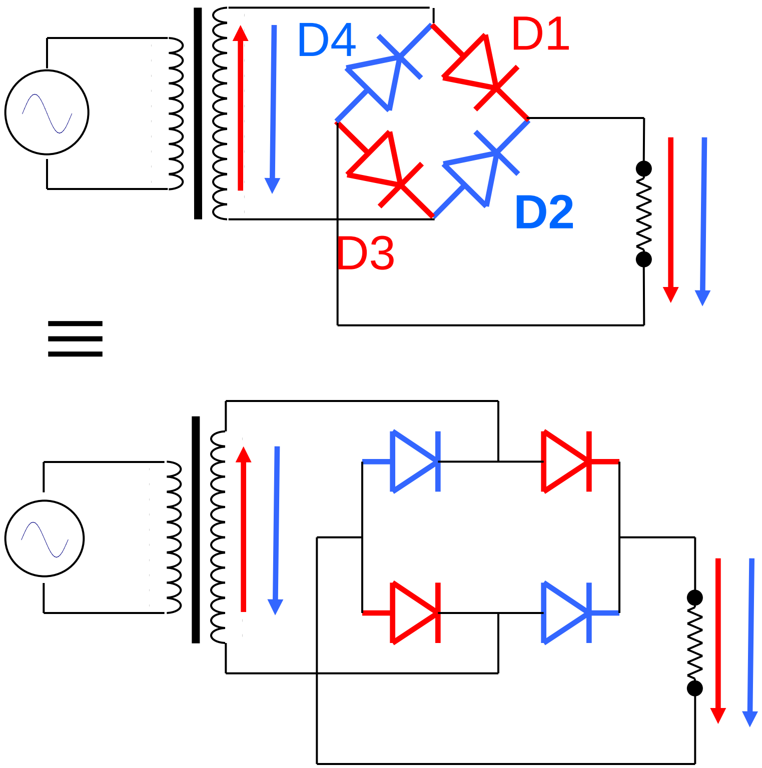


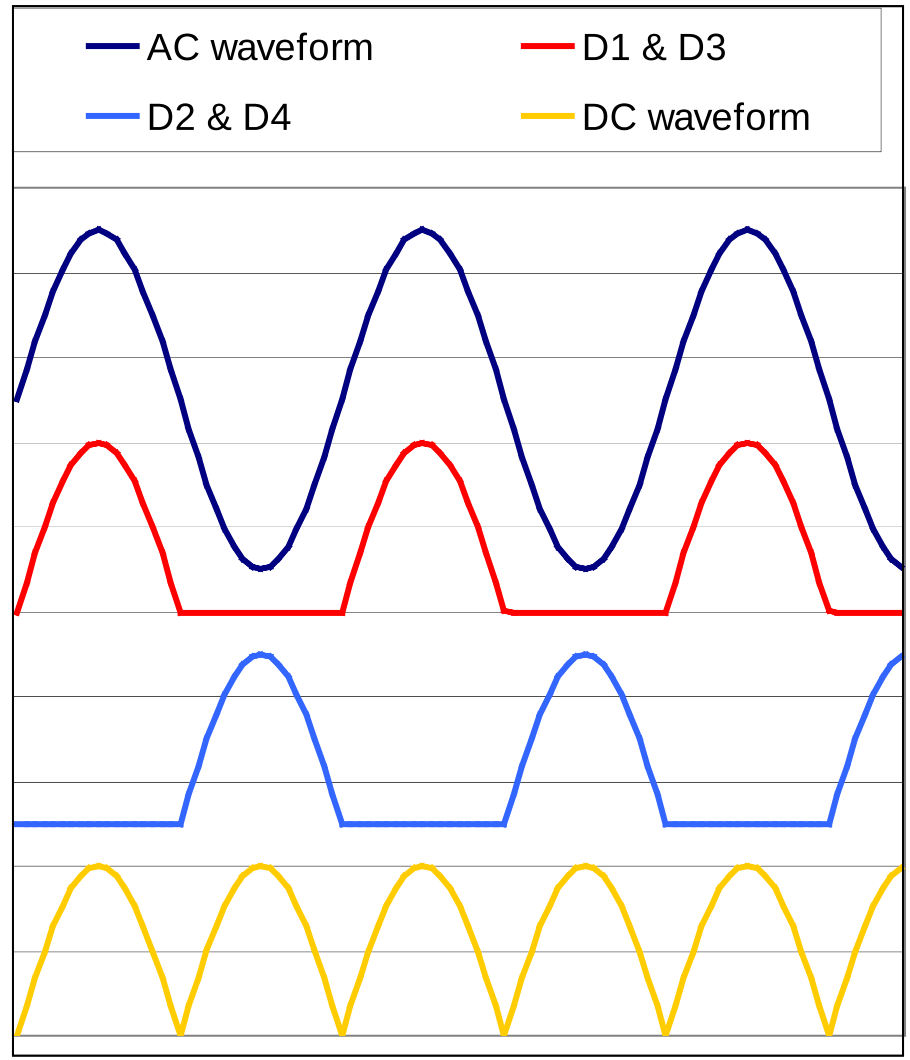
Half-wave case

Full-wave bridge rectifier circuits.

We can improve the previous circuit still further by the use of an additional pair of diodes. Now the transformer's secondary winding is fully utilised and hence a centre tap is not required. This is usually the most cost effective circuit to use.

Single phase bridges.





Note diodes all point the same way!

The effect of the form factor on the capacitor ripple voltage

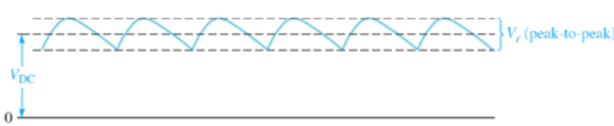
So far we have approximated the charging pulse in the capacitor to be of such a short time that it doesn't affect the voltage waveform. In reality, the charging pulse affects the waveform sufficiently to warrant its inclusion. Whilst the diode is conducting, it is supplying the load AND re-charging the capacitor. Hence the capacitor discharge period is shortened by the diode's conduction period.

Example. If the input to a full-wave rectifier is 12V @ 50Hz, R_{Load} is 8.5Ω and $C_{Smoothing}$ 10,000µF, what is the ripple voltage if the diode conduction angle is 30°?

f = 50Hz => full-wave rectified pulse freq is 100Hz. Period = 10ms, Conduction angle = 30° or 1.66mS, so capacitor discharge time is 8.33mS. Peak load voltage is √2 x 12V = $17V_{Peak}$. Load = 8.5Ω so load current is 2A.

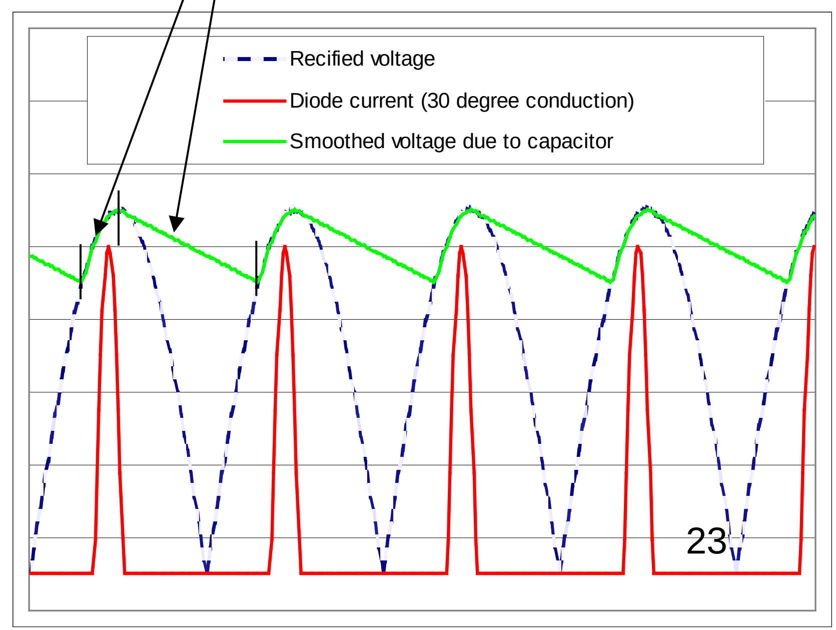
$$I_L = C\frac{dV_L}{dt} \quad or \quad dV_L = \frac{I_L.dt}{C} = \frac{2A \times 8.33mS}{10mF} = 1.66V$$

$$V_{ripple} = \frac{V_p}{R_L C} \times \Delta t$$



Power supplied by diode

Power supplied by capacitor



Notes.

Assume load current doesn't vary during discharge period. Approximate to a linear discharge ramp.

Peak to peak ripple is 1.66V and mean ripple is half this (0.83V) Mean output is 16.2V.

Larger cap: **higher mean output voltage, reduced ripple, decreased diode conduction angle, higher diode current**.

Example:

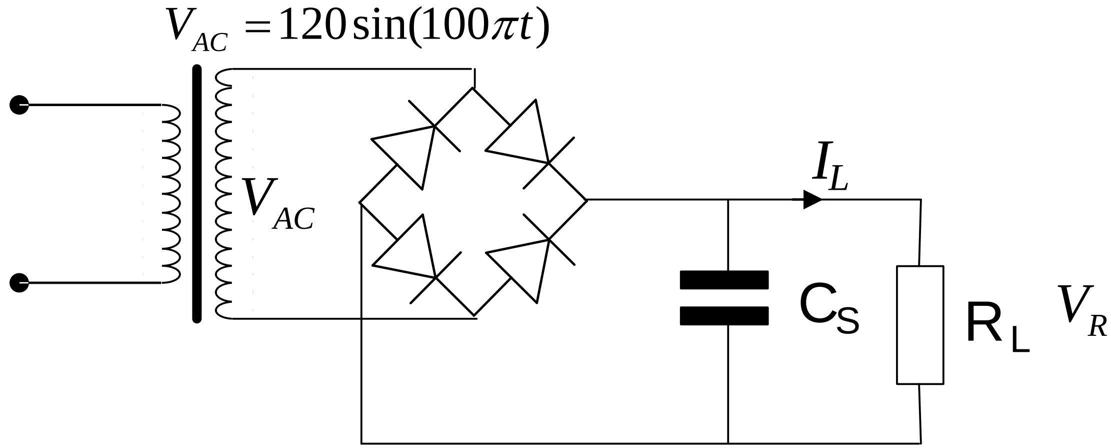
For the full-wave diode rectifier circuit shown in Figure below, assuming the input voltage is $V_{AC} = 120\sin(100\pi t)$ the conduction angle of the diode is 30°, and the load resistance R_L = 10Ω, and the capacitor C_S = 15,000μF. We assume that all components are perfect and neglect the voltage drop across the diodes.

(a) Estimate the peak-to-peak ripple voltage in the output voltage V_R and the peak inverse voltage in the voltage drop V_D for the diode D.

(b) If the output filter capacitor C_s is removed, sketch and label the voltages across the load R_L and the diode D separately, showing the time scale clearly. Calculate the RMS value of the output voltage V_R.

(c) If the ripple voltage across the load resistance R_L must not exceed 1V peak-to-peak. Calculate the value of smoothing capacitance required. You may assume that the capacitor provides current to the load for each complete cycle of the rectified voltage.

(d) Sketch and label the waveform of the voltage $V_R(t)$ across the load assuming that the smoothing capacitor has a very large capacitance.



Answers:

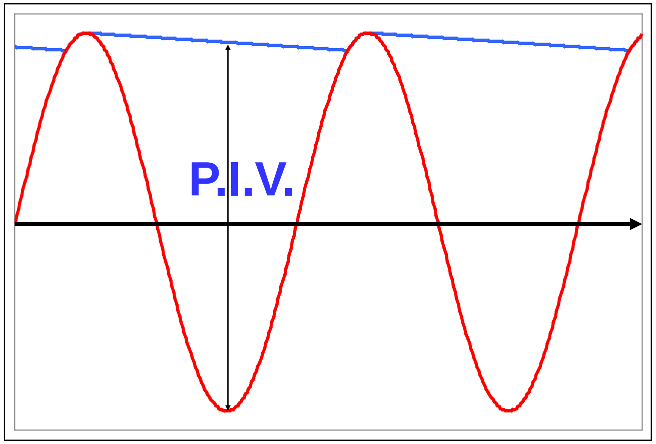
(a)The input voltage frequency is 50 Hz, i.e. the period at the output side is 10 ms, and the conduction angle of 30° means the conduction time is 30/180*10 ms=1.667ms, and the discharging time is Δt=10-1.667=8.333ms. Thus Peak load voltage is 120 V = 120V Peak. Load = 10 Ω so load current I_C is =120/10= 12A

The peak-peak voltage ripple is: $\Delta v_C = \frac{1}{C} \cdot I_C \cdot \Delta t = \frac{1}{0.015} \times 12 \times 0.008333 = 6.66\text{V}$

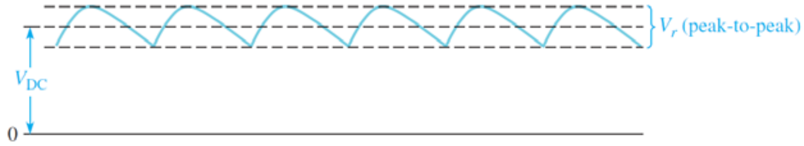
The peak inverse voltage drop V_D is: $V_{PIV} = 120\text{V}$

For half wave case PIV=(2$V_{p(sec)}$-V_{PP}/2)

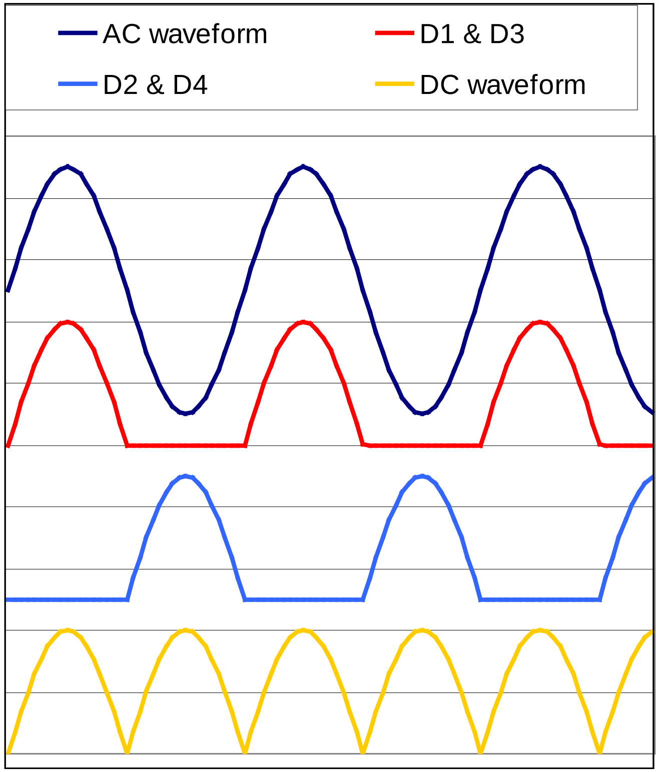
PIV=2$V_{p(sec)}$-V_{PP}/2



Half-wave case

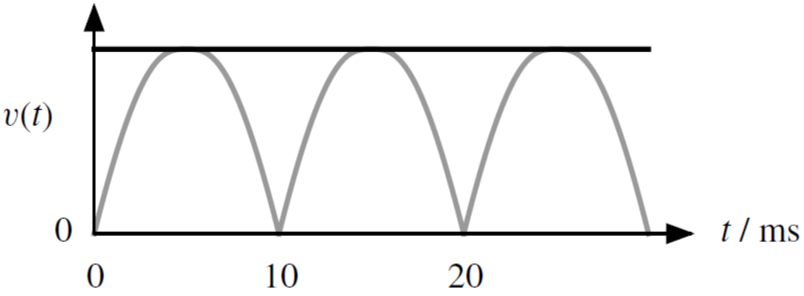


(b)



(c) $$\Delta v_C = \frac{1}{C}\cdot I_C \cdot \Delta t \le 1 \Rightarrow \frac{1}{C}\times 12\times 0.008333 \le 1 \Rightarrow C \ge 99996\mu F$$

(d)



Full bridge rectifier

Advantages:

(i) Full wave, easier to smooth

(ii)Uses transformer fully

(iii)Diodes need a peak inverse voltage (PIV) equal to the output voltage [not double as in some other circuits]

Disadvantages:

(i) Loses voltage from two diode drops

(i) Requires four diodes [not just one or two]

Two ways in which the ripple voltage had to be reduced further:

(i) Increase the value of the smoothing capacitor. This makes the current peaks more extreme, increasing the voltage drop (and power dissipation) across the transformer.

(ii) Add a linear regulator to the circuit. The input voltage must be increased to account for the regulator's dropout voltage.

Question

Design a power supply to deliver a current of 5A at 12V using a linear regulator. The regulator has a minimum input voltage of 14.5V and the mains input voltage is 230V +10% / - 6% @ 50Hz. The transformer has a regulation of 5% and is 93% efficient on full load. Rectification should be by the use of a bridge rectifier. Assume the diode conduction angle at full load is 36°.

Answer the following questions:

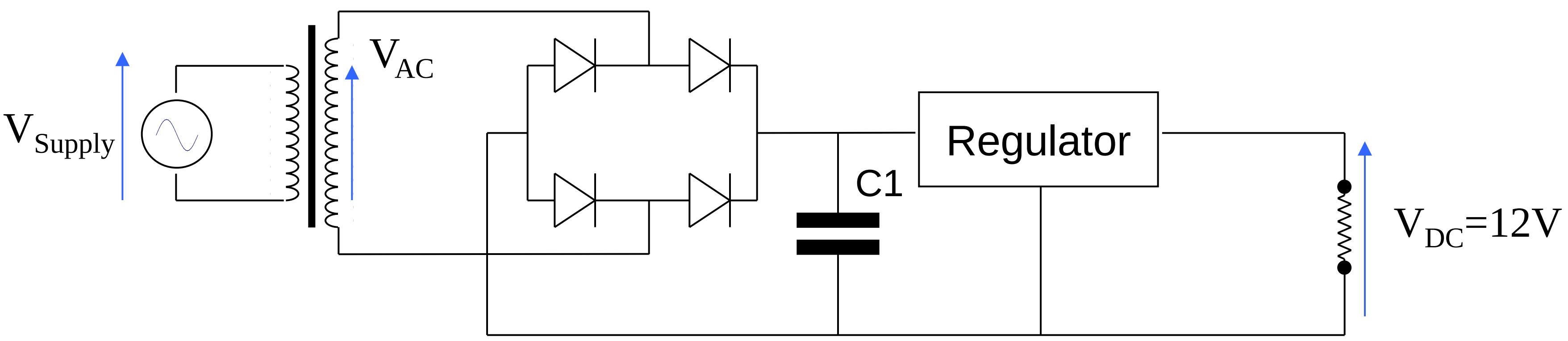
A. What is the minimum AC voltage the secondary winding should have if the smoothing capacitor is 10,000μF?

B. What is the power dissipation in the regulator under worst-case conditions?

C. What voltage rating should the capacitor have?

D. What PIV rating should the diodes have?

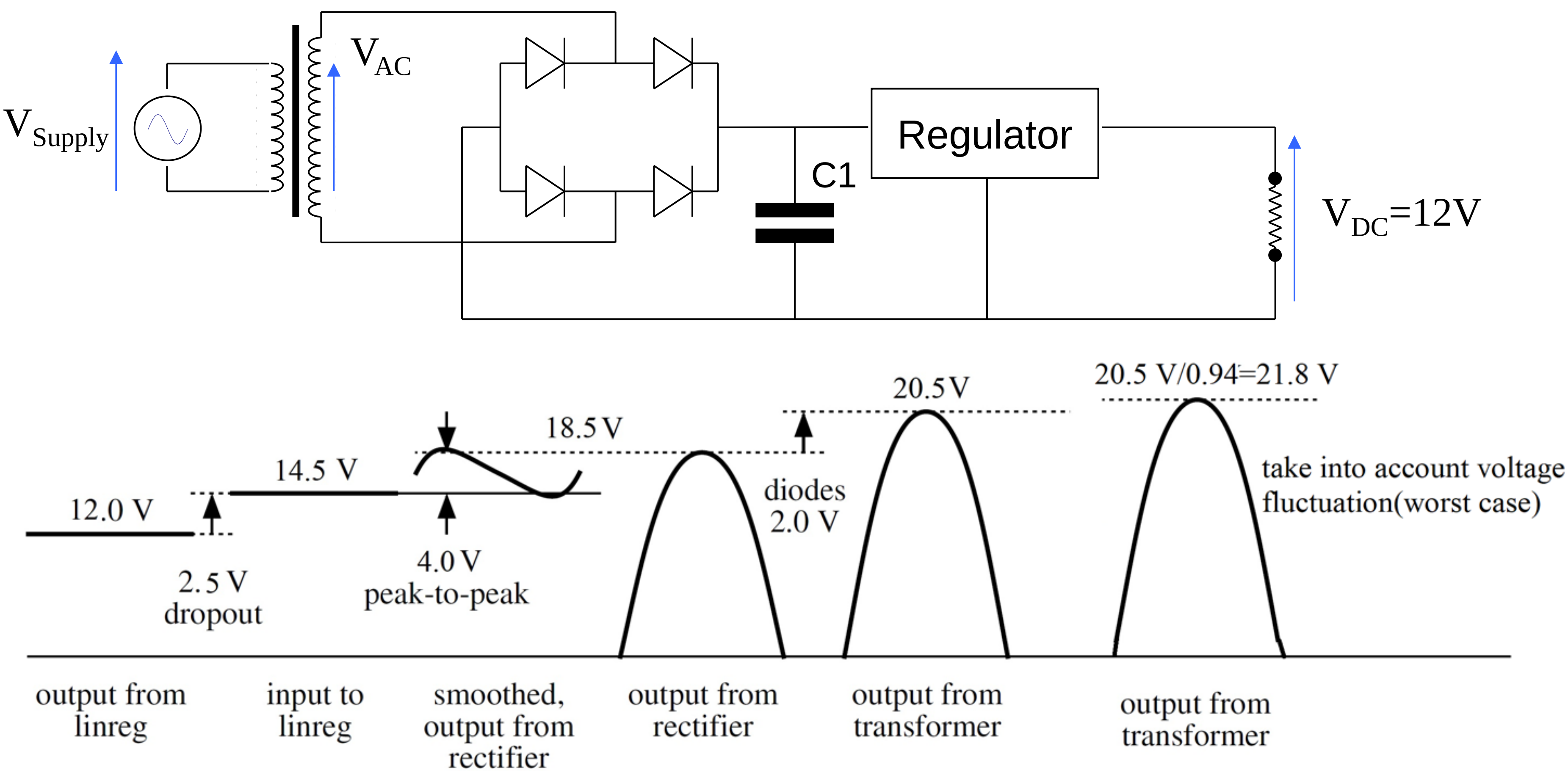
E. What VA rating does the transformer require?



Transformer regulation = 5%, η = 93%, V_{Supply} = 230V AC +10%, -6% @ 50Hz

C1 = 10,000μF, Regulator V_{IN} = 14.5V, I_{Max} = 5A, V_{OUT} = 12V

Voltage waveforms at different points in the power supply with linear regulator (linreg), working back from the output



$$V_{RMS} = \frac{21.8}{\sqrt{2}} = 15.3\text{ V} \approx 16\text{ V}$$

Solution to A

Start at the regulator and work back towards the mains supply.

Step 1. Find out what the minimum DC voltage we need to drive the regulator is.
The regulator output voltage is 12V and it needs an extra 2.5V "headroom". So not less than 14.5V input.

Step 2. Calculate the maximum ripple voltage (which depends on load current).
We know the maximum current is 5A and the capacitor is 10,000uF. We are also told the diode conduction angle is 36° (one fifth of the half cycle and the supply frequency is 50Hz. From this, we can calculate the peak to peak ripple voltage. We need to use the peak to peak rather than the average since the input to the regulator must ***always*** be kept above 14.5V.

$$\text{Use } \mathrm{I} = \mathrm{C}\frac{\mathrm{dV}}{\mathrm{dt}}:\ V_{RIPPLE} = \frac{I.\Delta t}{C} = \frac{5\times(10-2mS)}{10mF} = 4V$$

Thus the minimum value of the rectified voltage seen at the capacitor must be at least 18.5V to ensure the ripple voltage doesn't cause the regulator input to fall below 14.5V.

Step 3. Don't forget to add the voltage drop across the diodes. When the supply has no output current the voltage drop across each diode will be 0V. However, on load, assume each diode voltage drop is 1V.
Since we're using a bridge rectifier then there are two diodes in series (one in the +ve side & one in the –ve side). Thus the peak voltage needed is 18.5V + 2V => V_{Peak} = 20.5V .

Step 4. We also need to take into account the variation in the supply voltage. The nominal is 230V but it can drop down as low as 230V(1 – 6%). At this level, we still need to ensure the peak voltage is ≥ 20.5V. Hence the peak secondary voltage needs to be divided by 0.94. So 20.5V minimum now becomes 20.5V / 0.94 = 21.8V.

Step 5. We need to express this peak voltage requirement in terms of the nominal AC RMS secondary voltage since this is what transformer manufacturers (and everybody else for that matter) actually use. We worked this out for the lowest supply voltage (the worst-case condition), so we just need to convert to RMS: divide the peak by √2 – so 21.8V_{peak} becomes 15.3V_{RMS}. This is the required voltage rating of the secondary winding of the transformer. In reality you would round it up to 16V AC.

Finally, also remember that the transformer's rated output voltage is delivered at maximum load, so don't include the effect of regulation for this calculation.

Solution to B - Power calculation

The worst case power dissipation in the voltage regulator will occur when it is delivering maximum output current and the mains supply voltage is at its highest value.

The highest supply voltage is 230V + 10%, and this corresponds to a 10% increase in V_{AC} we just worked out. Since we specified a 16V transformer, use this figure.

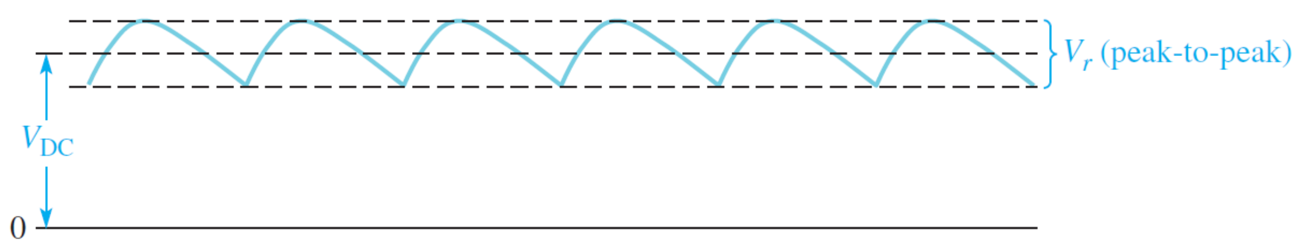
We need to calculate the average voltage drop across the regulator to find the power dissipated. We know the peak to peak ripple voltage is 4V, so the average ripple voltage is half this value: 2V. We need to subtract the voltage drop across the diodes, too. Hence the average input voltage to the regulator is:

$$V_{AC(Max)} = \frac{230V \times (10\% \; +1)}{230V} \times 16V = 17.6\, V_{RMS}$$

$$V_{REG\ INPUT} = (17.6 \times \sqrt{2}) - V_{DIODES} - Average\ ripple(2V) = 20.895V$$

$$I = 5A \Rightarrow P = I\Delta V = 5 \times (20.89 - 12V) \boxed{= 44.45W}$$

The power dissipated is the product of volts dropped across the regulator x the current flowing through it.



Solution to C & D – Highest voltages

The maximum capacitor voltage will be when there is no load on the power supply (i.e. the transformer regulation causes the output voltage to rise) and the supply voltage is at its highest : 230V + 10%. Also, at no load, the voltage drop across the diodes falls to zero (since no current is flowing).

$$V_{CAP(Max)} = V_{AC} \times \frac{230V \times (1+10\%)}{230V} \times 1.05 \times \sqrt{2} = \boxed{26.1V}$$

This is what voltage rating the capacitor needs.

The lowest value of the secondary voltage is negative of this value, hence $PIV = V_{CAP\,(Max)} - V_{AC\,(Peak)} = 52.2V$

However, for a bridge rectifier, there are two diodes in series, so the PIV rating for the diodes in this case is the same as for the capacitor since both diodes share the peak voltage. The peak inverse voltage across the diodes will occur when the capacitor voltage is at the highest value and the secondary voltage is at its lowest value in the sinusoid.

Diode voltage rating too.

Solution to E - calculation of VA rating.

A high form factor for the current waveform leads to additional heating in the transformer windings. To ensure the transformer temperature rise remains within the manufacturer's specification, we need to derate it. The working approximation is to say the AC current equals the DC current times the Form Factor of the waveform.

A 36° conduction angle per half cycle equates to 5ωt over π radians.

The following page has the mean and RMS calculations for 5ωt to find the form factor, 2.48 in this case .

We need to multiply the secondary VA load by the transformer's efficiency to get the total VA rating:

$$VA_{LOAD} = V_{AC} \times I_{AC} = V_{AC} \times F^* \times I_{DC} = 16 \times 2.48 \times 5 => VA_{LOAD} = 198.4VA$$

$$=> VA_{TRANSFORMER} = \frac{VA_{LOAD}}{\eta} = \frac{198.4}{0.93} = 213.3VA$$

Final remark: Note we have a transformer able to deliver over 200W into a resistive load and yet the linear power supply output is just 60W!

Average current

$$I_{ave(\alpha=36)} = \frac{1}{\pi}\left[\int_0^{\frac{\pi}{5}} \hat{I}\sin 5\omega t d\omega t + \int_{\frac{\pi}{5}}^{\pi} \hat{I}\sin 5\omega t d\omega t\right]$$

$$= \frac{\hat{I}}{\pi}\left[-\frac{1}{5}\cos 5\omega t\right]_0^{\frac{\pi}{5}} = \frac{-\hat{I}}{5\pi}[-2] = \boxed{\frac{2\hat{V}}{5\pi}}$$

Form factor

$$FF = \frac{I_{RMS}}{I_{AVE}} = \frac{\frac{\hat{I}}{\sqrt{10}}}{\frac{2\hat{I}}{5\pi}} = \frac{5\pi}{2\sqrt{10}} = \boxed{2.48}$$

RMS current

$$I^2_{RMS(\alpha=36)} = \frac{1}{\pi}\left[\int_0^{\pi/5}\left(\hat{I}\sin 5\omega t\right)^2 d\omega t + \int_{\frac{\pi}{5}}^{\pi} 0.d\omega t\right]$$

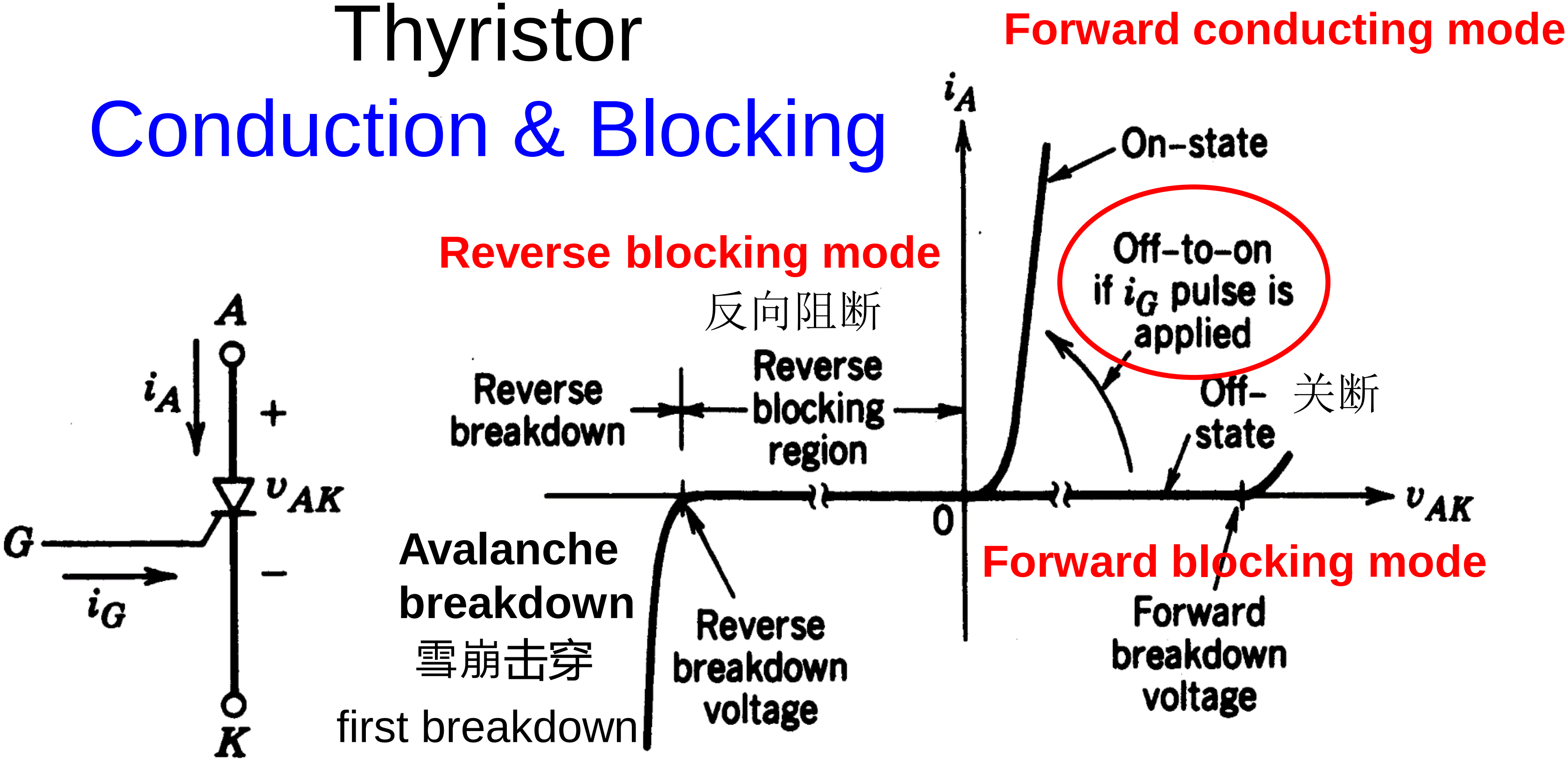
$$= \frac{\hat{I}^2}{\pi}\int_0^{\frac{\pi}{5}} \sin^2 5\omega t d\omega t = \frac{\hat{I}^2}{\pi}\int_0^{\frac{\pi}{5}} \frac{1}{2}[1-\cos 10\omega t]$$

$$= \frac{\hat{I}^2}{2\pi}\left[[\omega t]_0^{\pi/5} + \left[\frac{1}{10}\sin 10\omega t\right]_0^{\pi/5}\right] = \frac{\hat{I}^2}{2\pi}\times\frac{\pi}{5} = \frac{I^2}{10}$$

$$\Rightarrow I_{RMS(\alpha=36)} = \boxed{\frac{\hat{I}}{\sqrt{10}}}$$

Thyristor

Conduction & Blocking



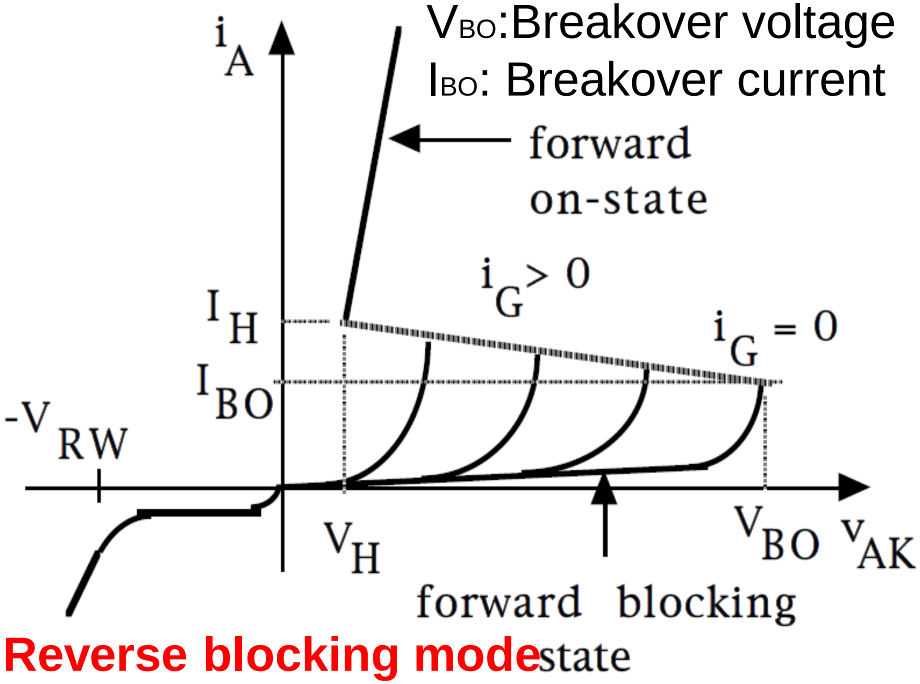
Half-Controllable Switch 半控开关

The thyristor can be turned on by applying a pulse of positive gate current i_G for a short duration when the device in its forward-blocking 前向阻断 state.

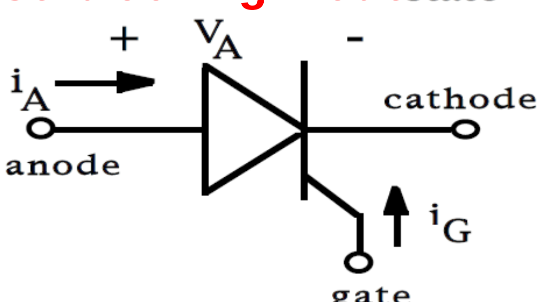
Once the device begins to conduct, it is latched on and I_G can be removed. It cannot be turned off by gate. It will turn off until anode current i_A go to zero.

Trigger On

V_{BO}:Breakover voltage
I_{BO}: Breakover current



Reverse blocking mode



Current to several kiloamps for V(on) of about 1 volts.

Blocking voltages to 5-8 kilovolts.

Maximum junction temperature = 125 °C

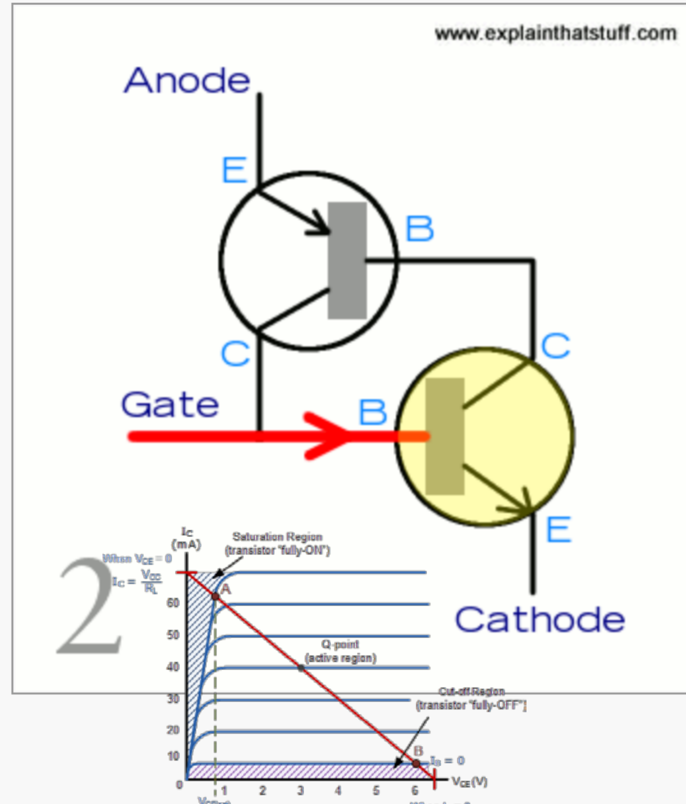
Higher i_G leads to shorter turn-on time

High efficient for high voltage high current applications (e.g. HVDC)

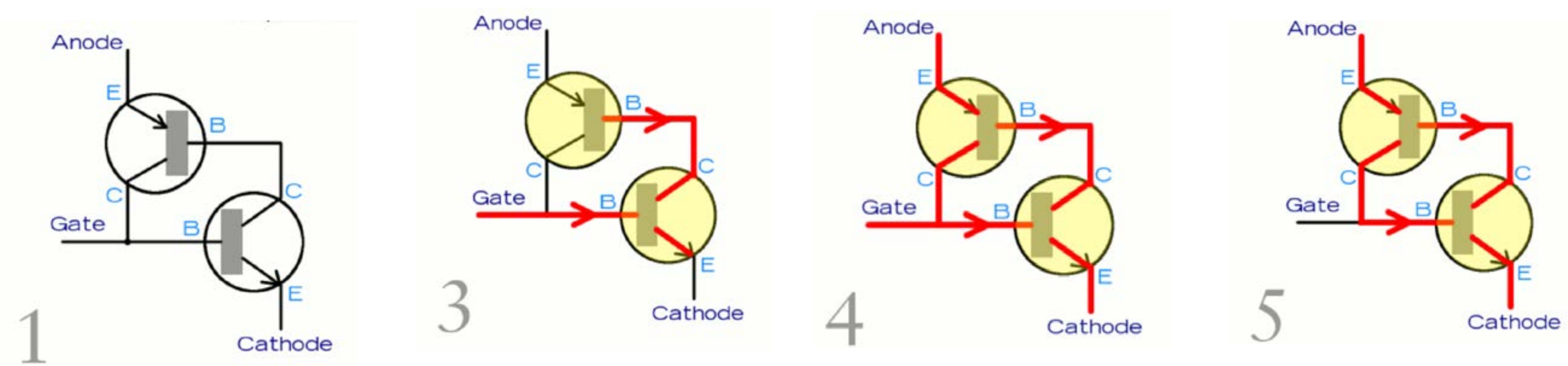
1. **Reverse blocking mode** – Voltage is applied in the direction that would be blocked by a diode
2. **Forward blocking mode** – Voltage is applied in the direction that would cause a diode to conduct, but the thyristor has not been triggered into conduction
3. **Forward conducting mode** – The thyristor has been triggered into conduction and will remain conducting until the forward current drops below a threshold value known as the "holding current" I_H which corresponds the holding voltage V_H

How a thyristor latches on

This little animation is a simple summary of how a thyristor latches on. You'll notice I've redrawn the thyristor so it looks like two transistors (a p-n-p at the top and an n-p-n beneath it) connected together, with the anode, cathode, and gate making the three external connections. Each transistor acts as the input to the other. So how does it work?

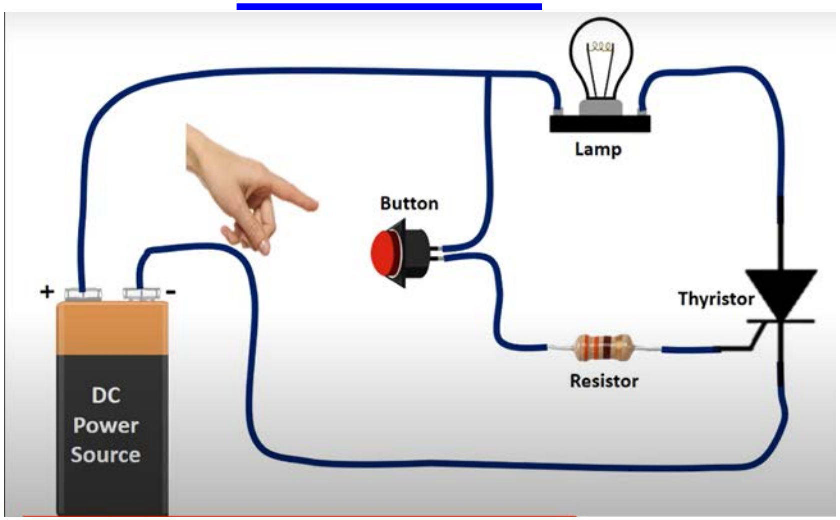


1. With no current flowing into the gate, the thyristor is switched off and no current flows between the anode and the cathode.
2. When a current flows into the gate, it effectively flows into the base (input) of the lower (n-p-n) transistor, turning it on.
3. Once the lower transistor is switched on, current can flow through it, activating the base (input) of the upper (p-n-p) transistor, turning that on as well.
4. Once both transistors are turned on completely ("saturated"), current can flow all the way through both of them—through the entire thyristor from the anode to the cathode.
5. Since the two transistors keep one another switched on, the thyristor stays on—"latches"—even if the gate current is removed.

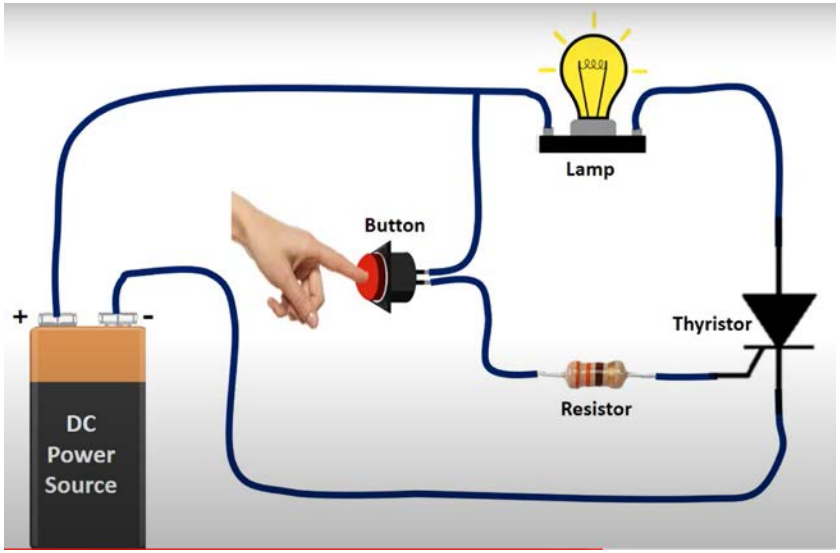


How thyristor works

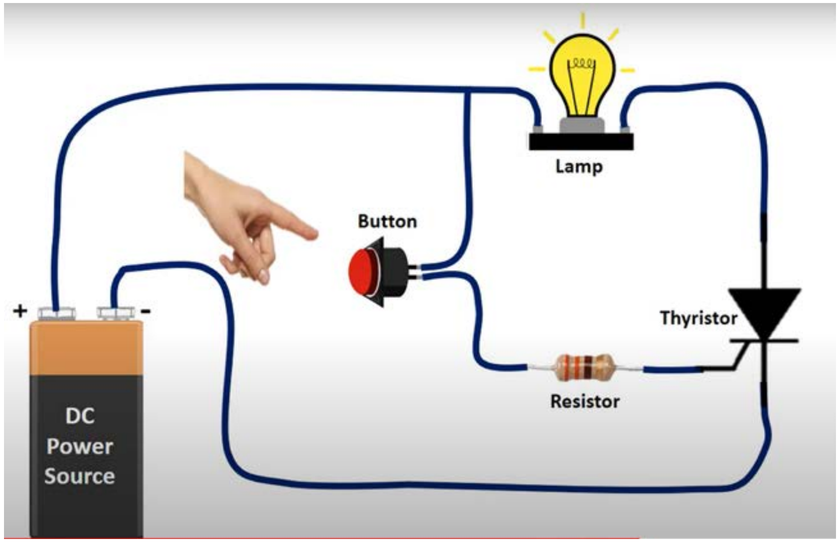
Button off

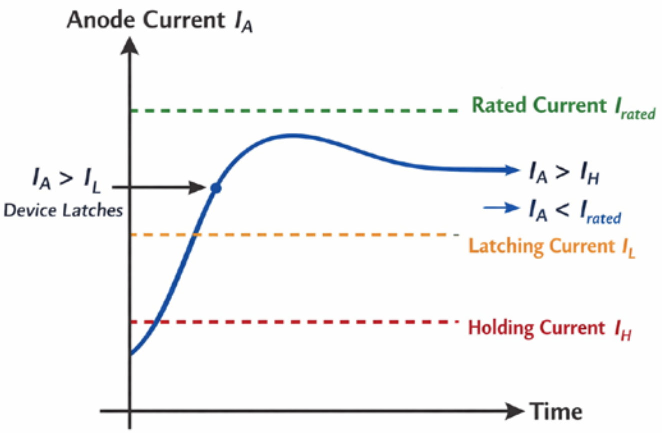


Button on



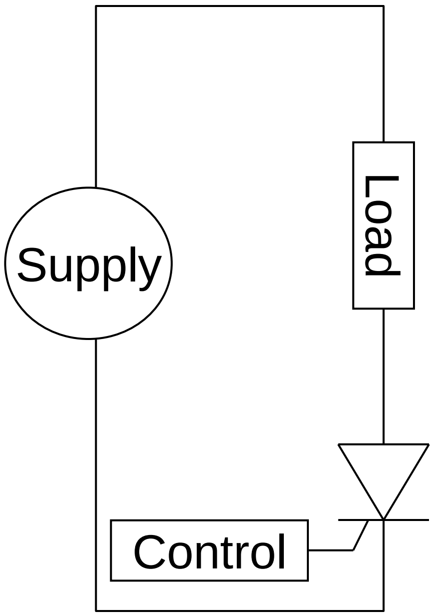
Button off again

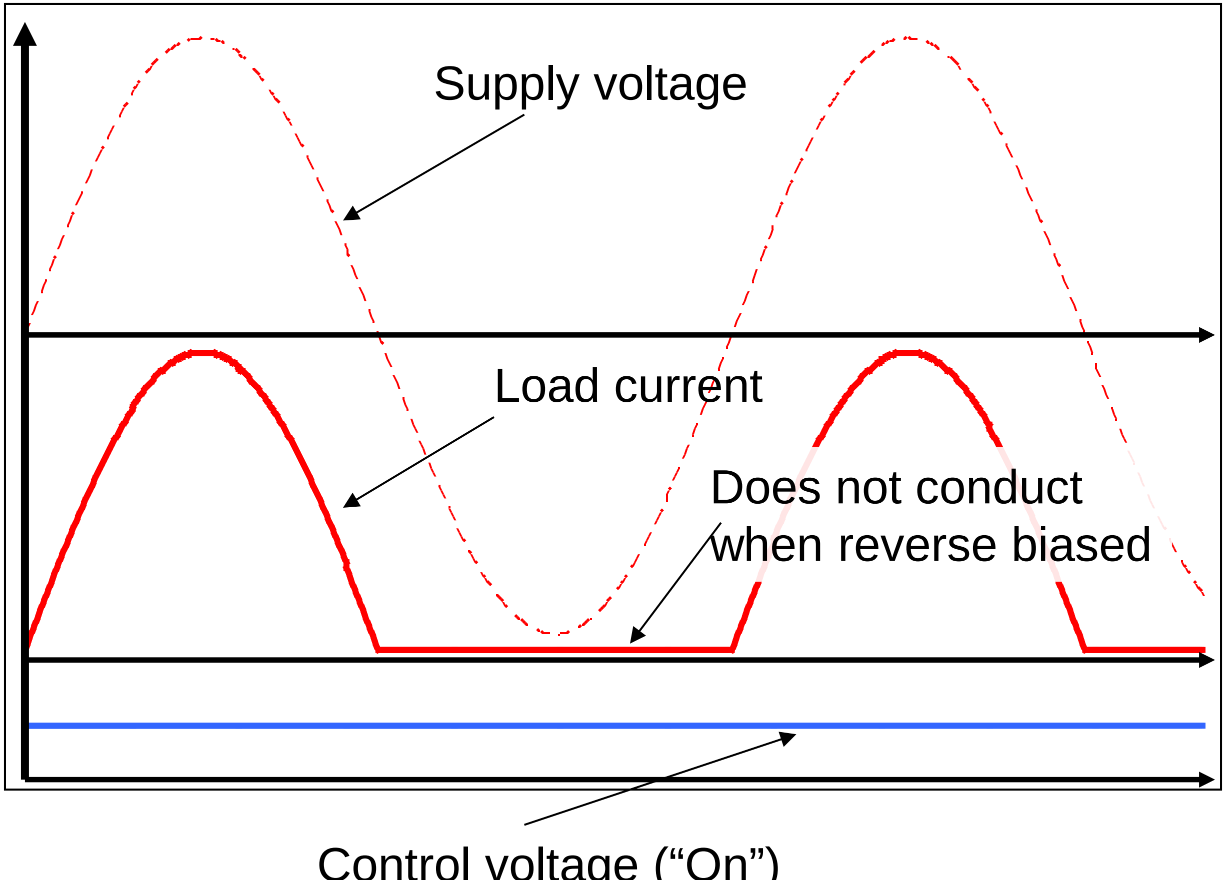




Thyrisitor Switching

The previous showed how an SCR can be used as switch in a DC circuit to protect the equipment connected to the output of a power supply. This took advantage of the fact that once an SCR is on, it stays on until the current through it falls to zero. We would also like to be able to use SCRs in AC circuits because of their huge power control capacity.



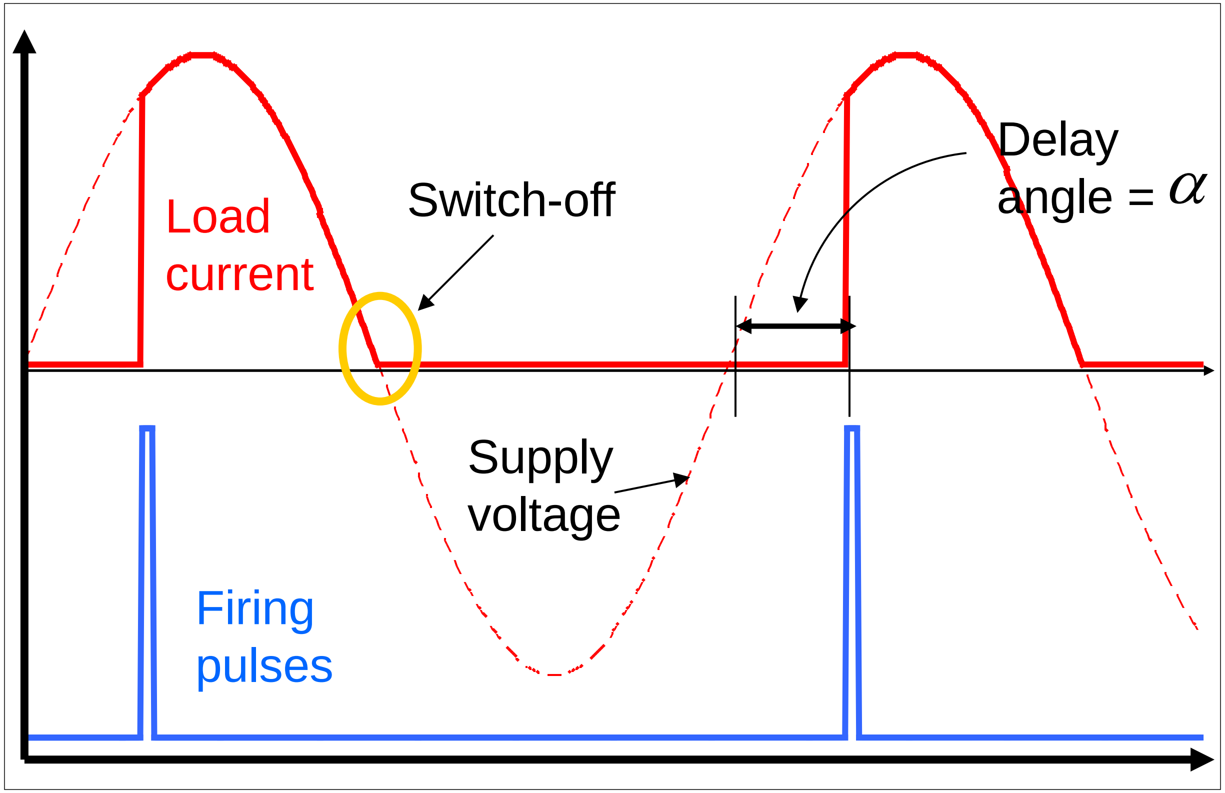


Thyristor behaving as a half-wave rectifier. The SCR turns off (“commutates”换流) automatically at the end of each positive half cycle.

SCRs are unsuitable for use in DC-DC conversion, because DC voltage cannot provide negative voltage for the turn-off of the switch

Phase control of SCRs

If we delay the firing pulse 触发脉冲 for the SCR relative to the zero-crossing points, we can use the SCR as a means 手段 of regulating the power applied to the load.



The period of time before the current flows in the SCR is called the delay angle, α

Calculation of average and RMS voltages for half-wave SCR phase angle control

$$V_{AVE} = \frac{1}{2\pi}\left[\int_0^{\alpha} 0.d\omega t + \int_{\alpha}^{\pi} \hat{V}\sin(\omega t)d\omega t\right]$$

$$= \frac{\hat{V}}{2\pi}\left[-\cos(\omega t\right]_{\alpha}^{\pi}$$

$$= \frac{\hat{V}}{2\pi}\left[-\cos(\pi) + \cos(\alpha)\right]$$

$$= \frac{\hat{V}}{2\pi}\left[1 + \cos(\alpha)\right]$$

$$V_{RMS}^2 = \frac{1}{2\pi}\left[\int_0^{\alpha} 0.d\omega t + \int_{\alpha}^{\pi} \left(\hat{V}\sin(\omega t)\right)^2 d\omega t\right]$$

$$= \frac{\hat{V}^2}{2\pi}\int_{\alpha}^{\pi}\left[\frac{1}{2}\left(1 - \cos(2\omega t)\right)d\omega t\right]$$

$$= \frac{\hat{V}^2}{4\pi}\left[\omega t - \frac{1}{2}\sin(2\omega t)\right]_{\alpha}^{\pi}$$

$$= \frac{\hat{V}^2}{4\pi}\left[\pi - \alpha + \frac{1}{2}\sin(2\alpha)\right]$$

Examples

For V_{MAX} = 100V

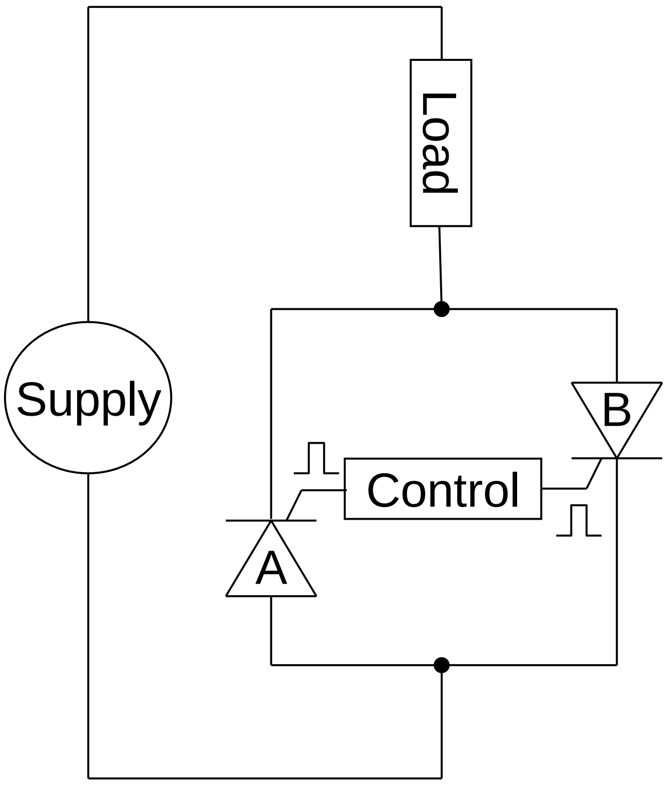
α= 45°: V_{AVE} = 27.17V, V_{RMS} = 47.67V

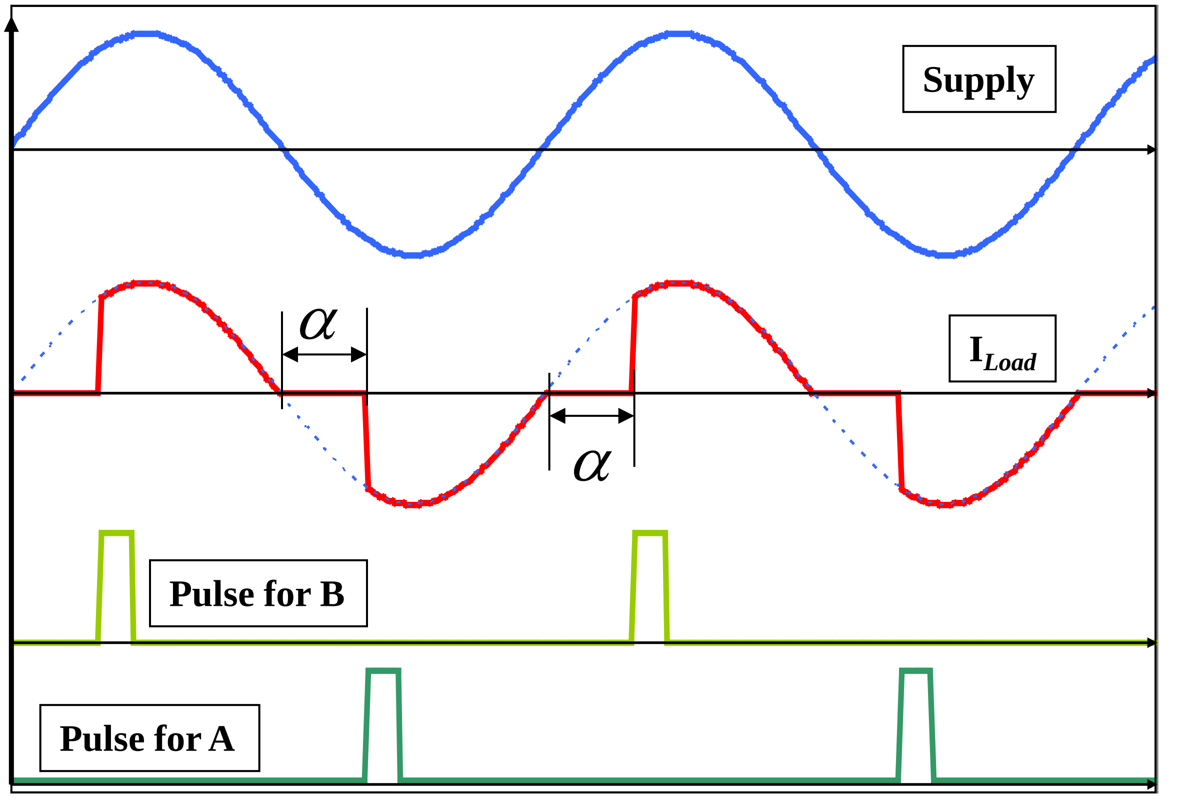
α= 90°: V_{AVE} = 15.92V, V_{RMS} = 35.36V

α= 135°: V_{AVE} = 4.66V, V_{RMS} = 15.07V

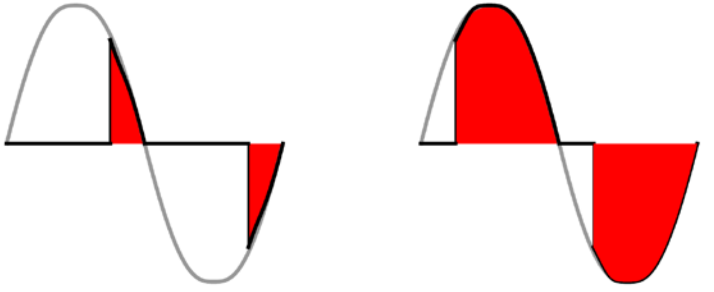
Back to back SCRs for full wave control

If we want to be able to control both halves of the mains waveform with SCRs we need to use a pair of SCRs.



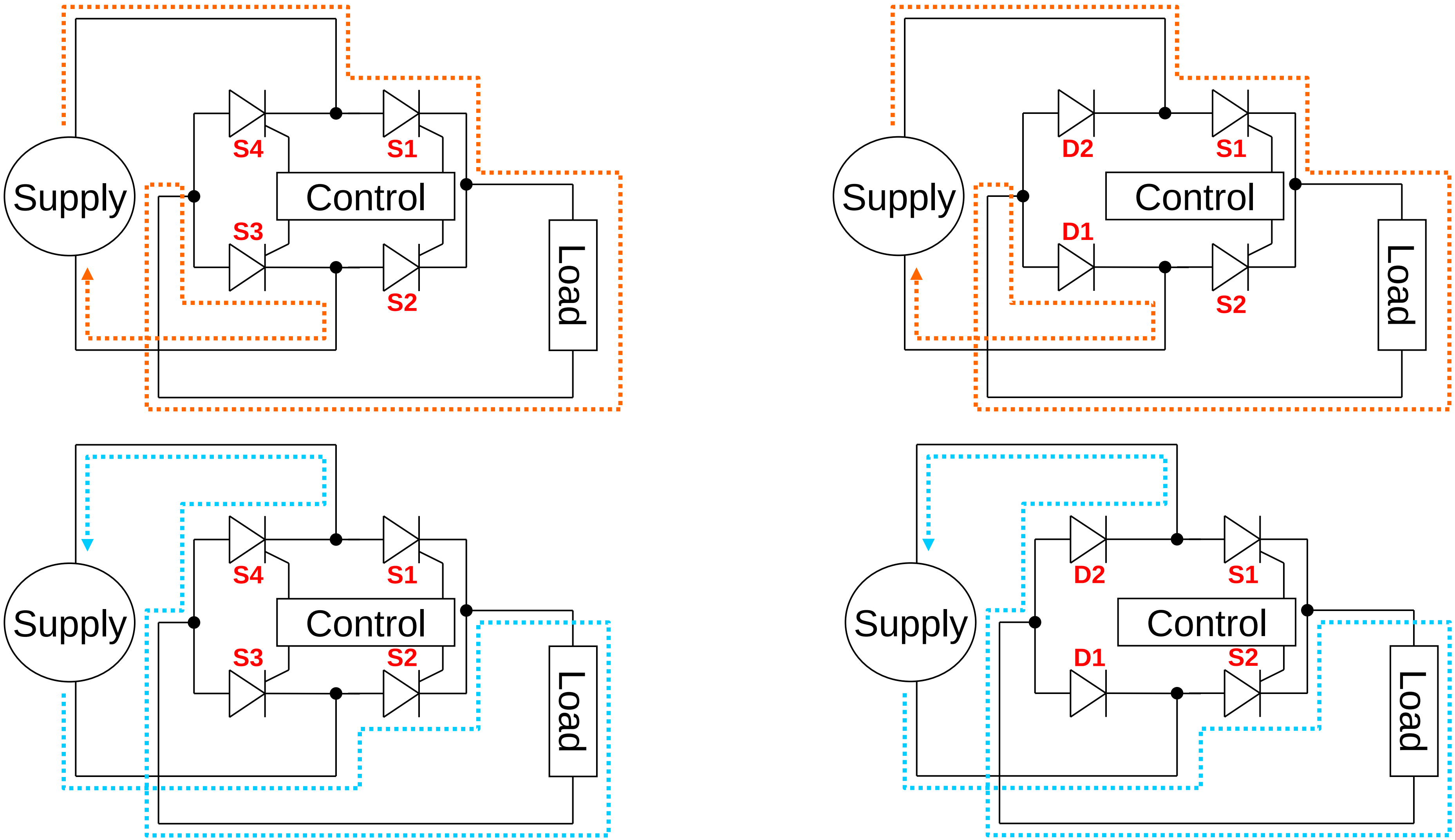


By varying α we can change the firing point and therefore vary the power in the load.



Thyristor Bridges

We can replace the diodes in a bridge rectifier with a pair of (or four) SCRs:

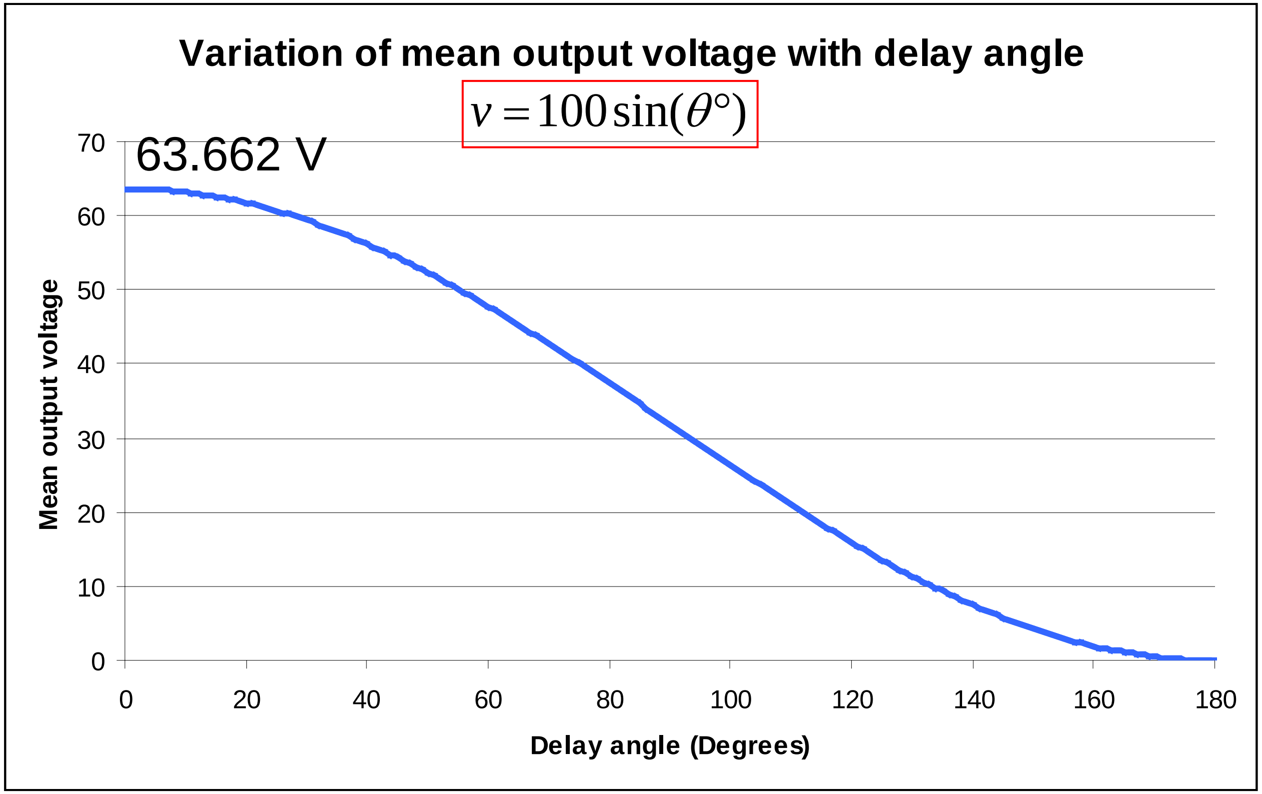


Positive & negative half-cycle current flows through SCRs and diodes.

When we introduce phase control into the bridge, the average voltage is given by the integral of the waveform from the delay angle to the end of the half-cycle.

$$\boxed{V_{AVE} = \frac{\hat{V}}{\pi}\left[1+\cos(\alpha)\right] \; ; 0 \le \alpha \le \pi}$$

The thyristor bridge allows us to vary the mean DC output voltage directly. The following chart of delay angle vs mean DC voltage is for a *resistive load only.*



$$V_{abs,ave} = \frac{\sqrt{2}V_{AC}}{\pi}\left(1+\cos\alpha\right)$$

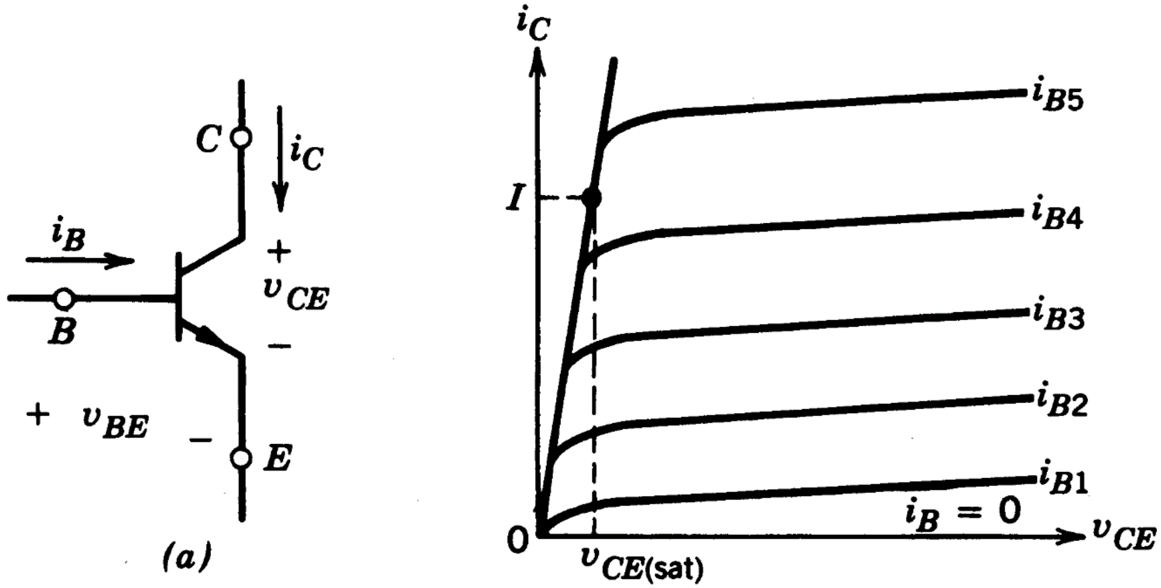
$\alpha = 90^\circ$, $\cos\alpha = 0$

$$V_{abs,ave} = \sqrt{2}V_{AC}\big/\pi = \sqrt{2}\times 230/\pi = 103.54 \text{ V}$$

$\alpha = 0$, $\cos\alpha = 1$

$$V_{abs,ave} = \frac{\sqrt{2}V_{AC}}{\pi}\left(1+\cos\alpha\right) = \frac{\sqrt{2}\times 230}{\pi}\left(1+1\right) = 207 \text{ V}$$

Bipolar Junction Transistor (BJT or GTR)



Now GTRs are replaced by MOSFETs, IGBTs

On-State Voltage: 1~2V

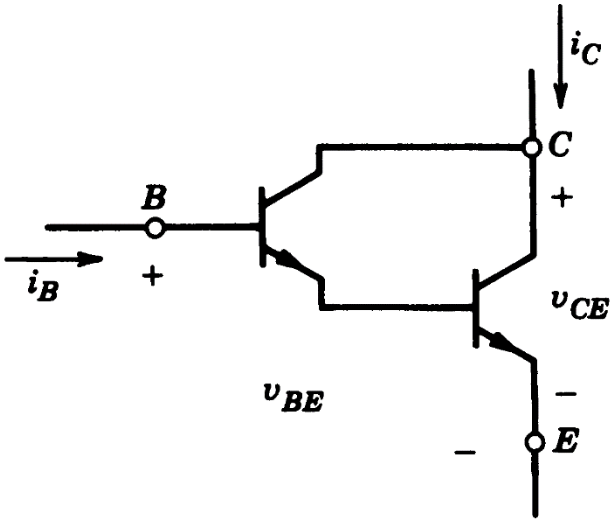
Limited Voltage/Current Rating: 1400V/several hundred A

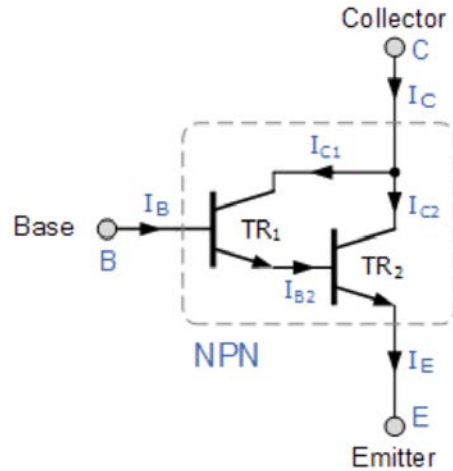
Limited Switching Frequency: several kHz

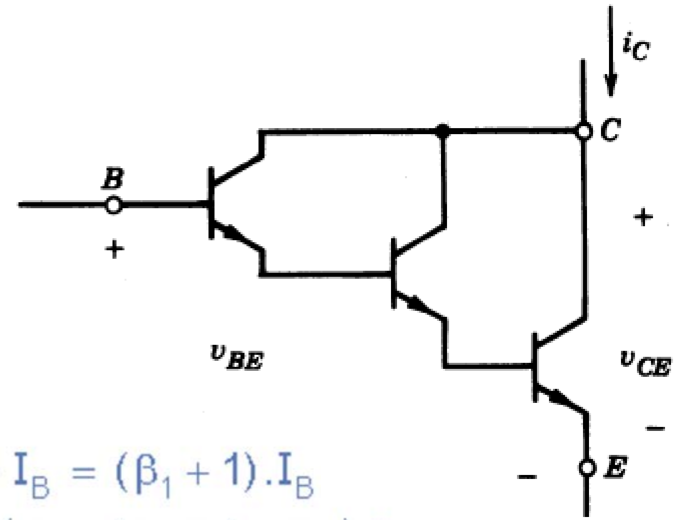
Complex current controlled drive circuit.

Darlington Configuration GTR

$$\boxed{I_B > \frac{I_C}{h_{FE}}}$$







$$I_{B2} = I_{E1} = I_{C1} + I_B = \beta_1.I_B + I_B = (\beta_1 + 1).I_B$$

$$I_C = I_{C1} + I_{C2}$$

$$I_C = \beta_1.I_B + \beta_2.I_{B2}$$

$$I_C = \beta_1.I_B + \beta_2.(\beta_1 + 1).I_B$$

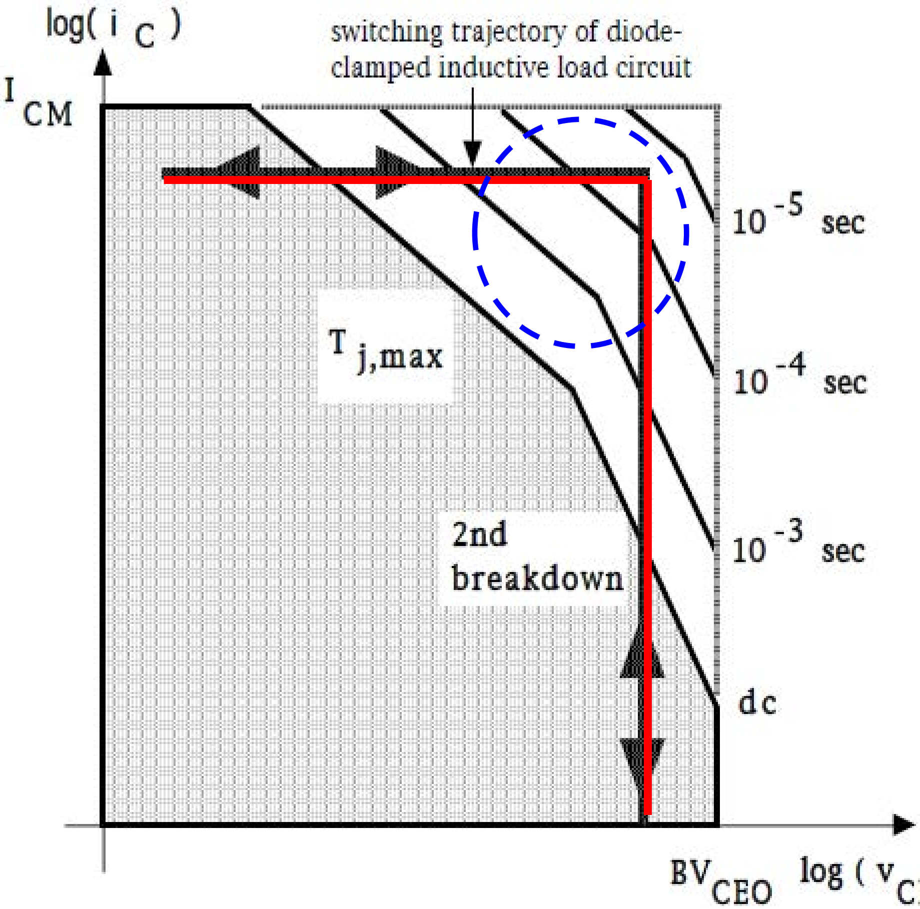
$$I_C = \left(\beta_1 + (\beta_2.\beta_1) + \beta_2\right).I_B$$

$$I_C = \beta_1.I_B + \beta_2.\beta_1.I_B + \beta_2.I_B$$

$$\beta_{\text{Darlington}} = \beta_1 \cdot \beta_2 + \beta_1 + \beta_2$$

BJTs are current controlled (fully-controllable) devices, base current i_B must be supplied continuously to keep them in the on state. To increase the dc current gain h_{FE} and reduce the gate current loss, BJTs usually adopt Darlington configuration达林顿结构 at cost of slightly higher $V_{CE(sat)}$ and slower switching speed. Driver circuit is also complex.

GTR Safe Operation Area
安全工作区



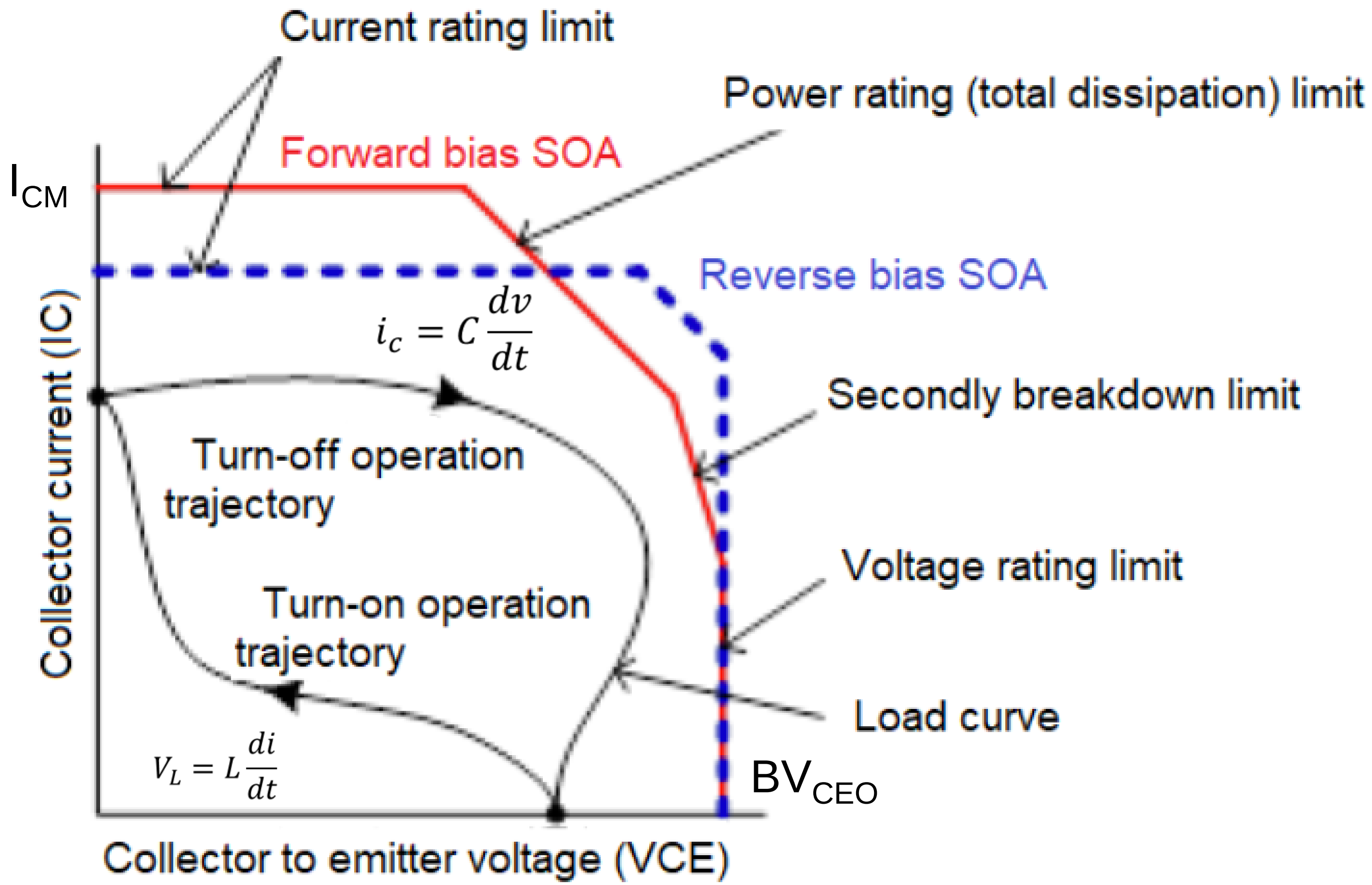
Forward bias 前置 safe operating area (FBSOA)

For power semiconductor devices (such as BJT, MOSFET, thyristor or IGBT), the safe operating area (SOA) is defined as the voltage and current conditions over which the device can be expected to operate without self-damage.

Snubber Circuit 缓冲电路 is needed for fully-controllable devices to shape its transient i_C-V_{CE} trajectory during the switching On/Off procedure.

Actually snubber is used to limit dv/dt and di/dt during switching transition.

Safe Operating Area(SOA)

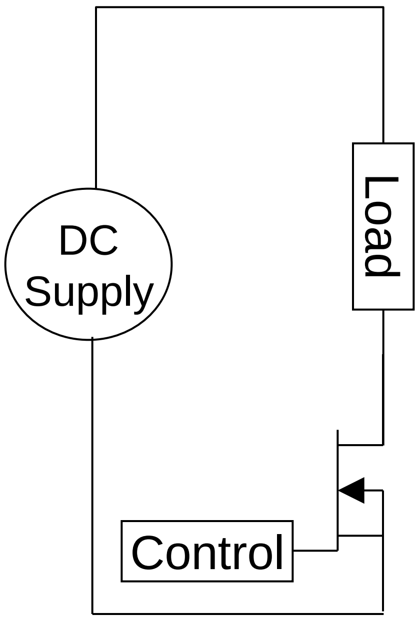


- In the datasheet, the company will supply the SOA picture to specify the maximum Ic , maximum junction temperature, the second breakdown curve, breakdown voltage limit BV_{CEO}

The MOSFET

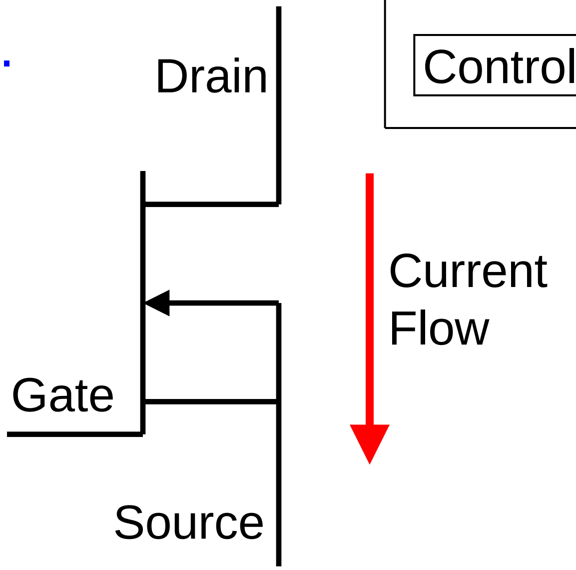
MOSFETs come in many varieties 型式 and sizes 尺寸. Some of their advantages include:

- Fast switching (10's → 100's nS).
- Huge power gain (Very simple gate control).
- Easily paralleled for higher current handling.



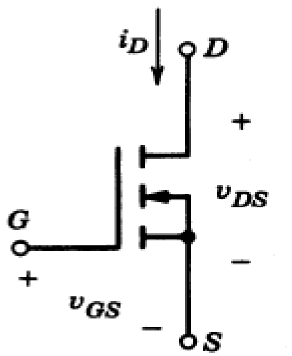
Their three main disadvantages are:

- Relatively low blocking voltage (1000V).
- Relatively low current handling (100A).
- Power dissipation can be high.

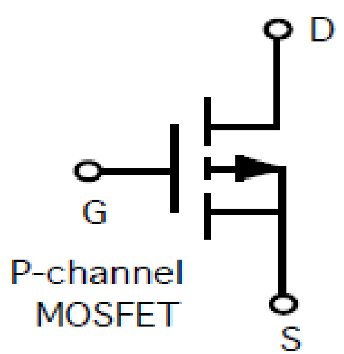


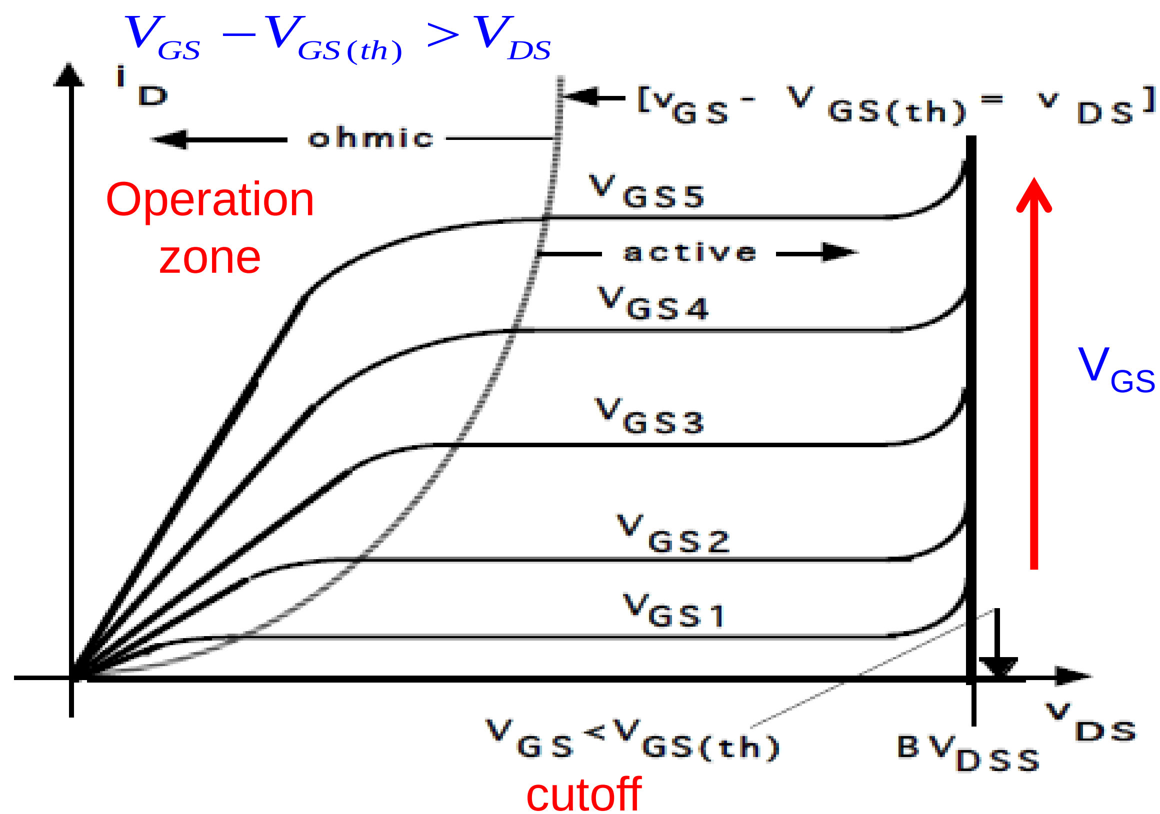
Metal Oxide Semiconductor Field-Effect Transistor (MOSFET)

金属氧化物半导体场效应管



N-channel





Voltage Controlled (Fully-Controllable) Device

Characteristics of MOSFET

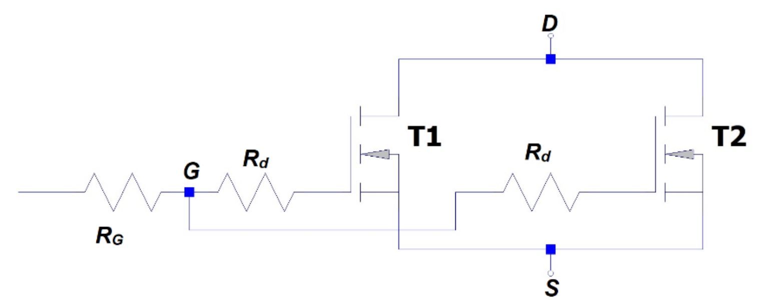
On-State Conduction Losses (Ohmic zone)

$$\boxed{p_{on} = I_d^2 r_{DS(on)}}$$

Good for synchronous rectification in low voltage applications.

$r_{DS(on)}$ is an approximated constant resistance

- Large V_{GS} minimizes accumulation layer resistance and channel resistance
- Positive temperature coefficient: $r_{DS(on)}$ increases as temperature increases. leads to thermal stabilization effect.
- MOSFETs can be easily paralleled due to positive temperature coefficient of $r_{DS(on)}$.

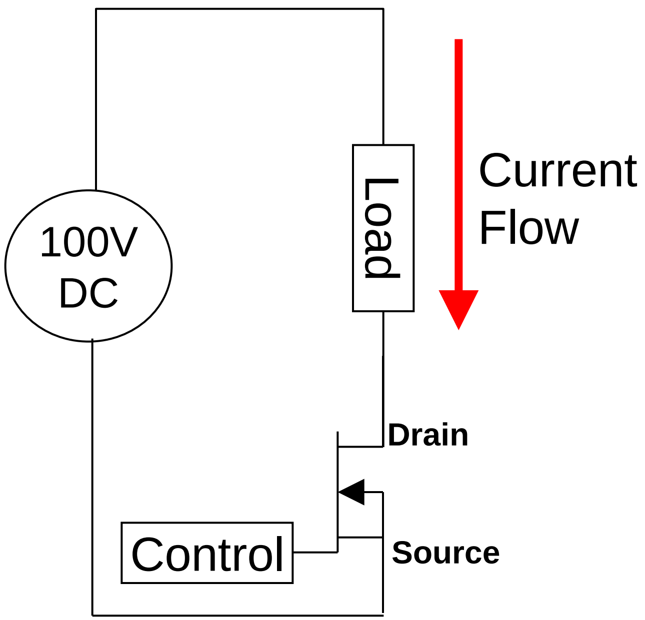


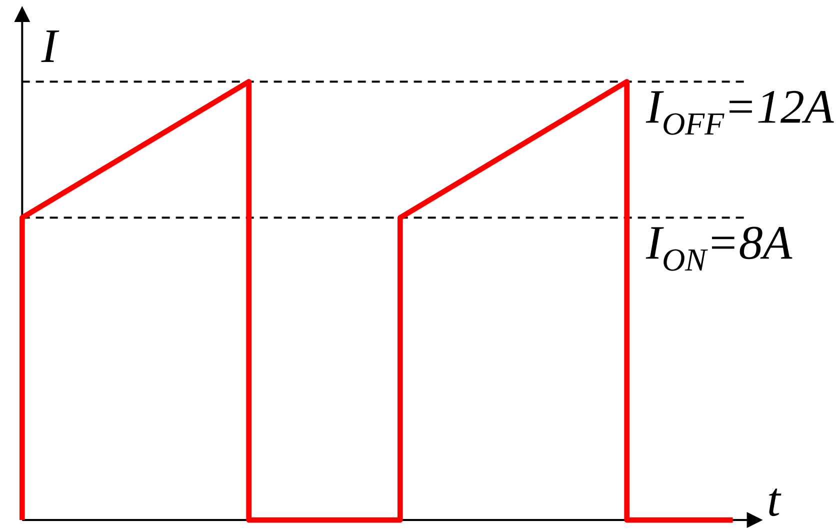
MOSFETs can be paralleled very easily, because of the positive temperature coefficient of their on-state resistance. For the same junction temperature, if $R_{DS(on)}$ of T_2 exceeds that of T_1, then during the on state, T_1 will have a higher current and thus higher power loss compared to T_2, since the same voltage appears across both transistors. Therefore, the junction temperature of T_1 will increase along with its on-state resistance. This will cause its share of current to decrease and, hence, there is a thermal stabilization effect

Example

Using the following data, determine the power loss in the MOSFET.

I_{ON}	8A
I_{OFF}	12A
PWM Duty	60%
PWM Freq	50KHz
Supply	100V DC
R_{ON}	50mΩ
T_{ON}	25nS
T_{OFF}	30nS





PWM period = 20μS
On time = 12μS
Off time = 8μS

Step 1 - Calculation of RMS current for the conduction loss.

Current function: $i(t) = I_{ON} + \frac{(I_{OFF} - I_{ON})t}{t_{ON}}; 0 < t \le t_{ON}$

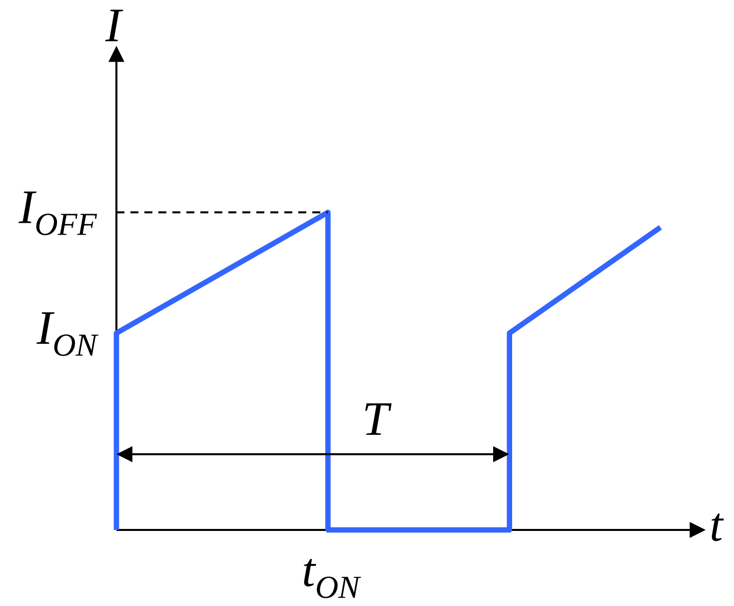
$$(i(t))^2 = I^2_{ON} + \frac{2I_{ON}(I_{OFF} - I_{ON})t}{t_{ON}} + \frac{(I_{OFF} - I_{ON})^2 t^2}{t^2_{ON}}$$

$$I^2_{RMS} = \frac{1}{T}\int_0^{t_{ON}} \left[I^2_{ON} + \frac{2I_{ON}(I_{OFF} - I_{ON})t}{t_{ON}} + \frac{(I_{OFF} - I_{ON})^2 t^2}{t^2_{ON}} \right] dt$$

$$= \frac{1}{T}\left[I^2_{ON}.t + \frac{I_{ON}(I_{OFF} - I_{ON})t^2}{t_{ON}} + \frac{1}{3}\cdot\frac{(I_{OFF} - I_{ON})^2 t^3}{t^2_{ON}} \right]_0^{t_{ON}}$$

$$= \frac{t_{ON}}{T}\left[I^2_{ON} + I_{OFF}.I_{ON} - I^2_{ON} + \frac{1}{3}\cdot\left(I^2_{OFF} - 2I_{OFF}.I_{ON} + I^2_{ON} \right) \right]$$

$$= \frac{\varphi}{3}\left[I^2_{OFF} + I_{OFF}.I_{ON} + I^2_{ON} \right]$$



For I_{ON} = 8A, I_{OFF} = 12A and ϕ = 60%, I_{RMS}=7.8A

$$\boxed{P_{Conduction} = I^2_{Load\ RMS} R_{DS\ ON}}$$

=> $P_{Conduction}$ = 7.8^2 x 50mΩ = 3.042W

Step 2 – calculation of switching loss.

$$P_{Switching} = \left(\frac{F_{SW} . V_{DS(Off)}}{2} \right) \left(T_{ON} . I_{ON} + T_{OFF} . I_{OFF} \right)$$

$$= \left(\frac{50KHz \times 100V}{2} \right) (25nS \times 8A + 30nS \times 12A)$$

$$= 1.4W$$

Step 3 – calculate total power loss

Power loss = Conduction loss + Switching loss

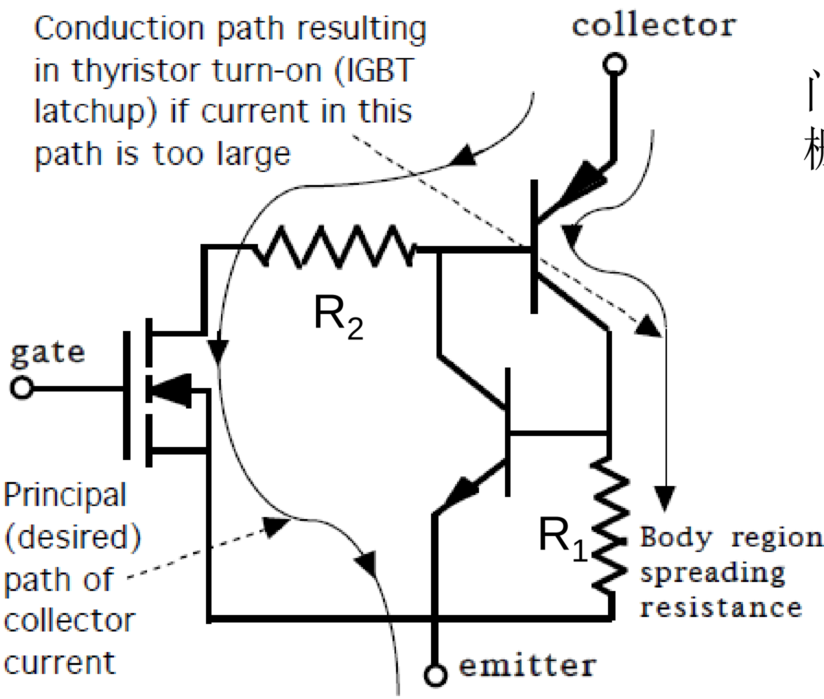
= 3.04W + 1.4W = 4.44W

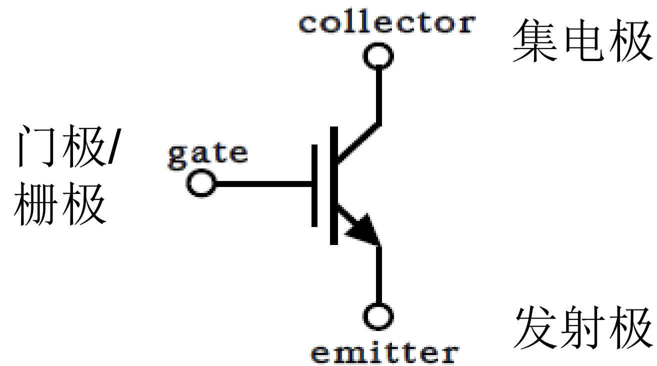
Step 4 – Calculate the average power consumption of the load

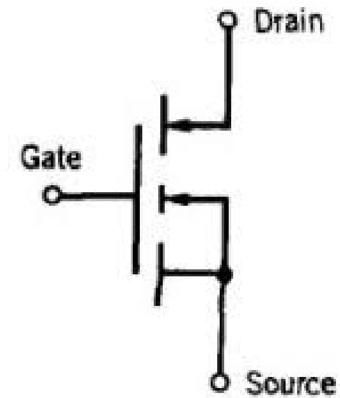
$P_{load} = I_{ave} \times V_{DC} = 6A \times 100V = 600$ W

Step 5 – Explain how the conduction and switching losses would change if the frequency of the waveform were increased by a factor of 10, assuming that the duty cycle and the shape of the current remained the same

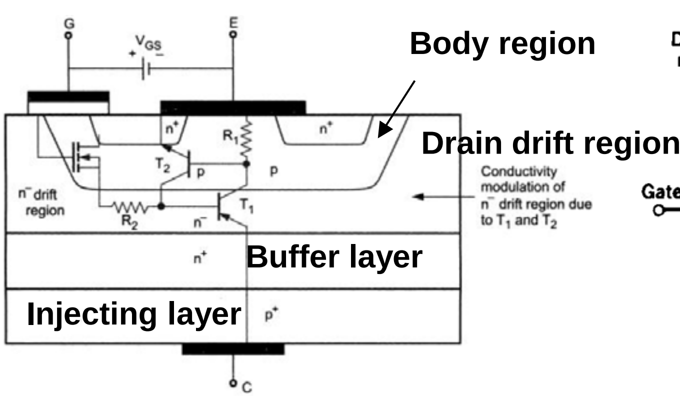
IGBT - Hybrid Switch

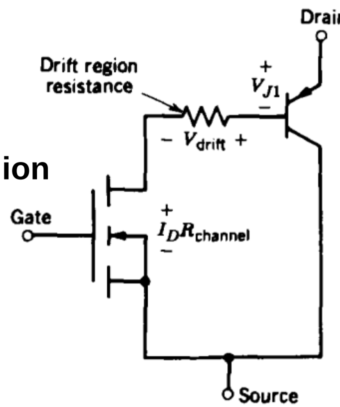






N-channel IGBT





N-channel IGBT Equivalent Circuit

$$V_{DS(on)} = V_{J1} + V_{drift} + I_D R_{channel}$$

- Gate Voltage V_{GE} Controlled Device like MOSFET
- Turn-on V_{CE} like GTR: 2~3V

IGBT principle and advantages over Power MOSFET

- **Operating Principle:**
- When a positive gate voltage ($V_{GE} > V_{th}$) is applied:
  - An inversion channel forms in the P-base region.
  - Electrons flow from emitter to drift region.
  - This triggers hole injection from the P+ collector.
  - The device conducts through bipolar action.
- When gate voltage is removed:
  - The channel disappears.
  - The device turns off.

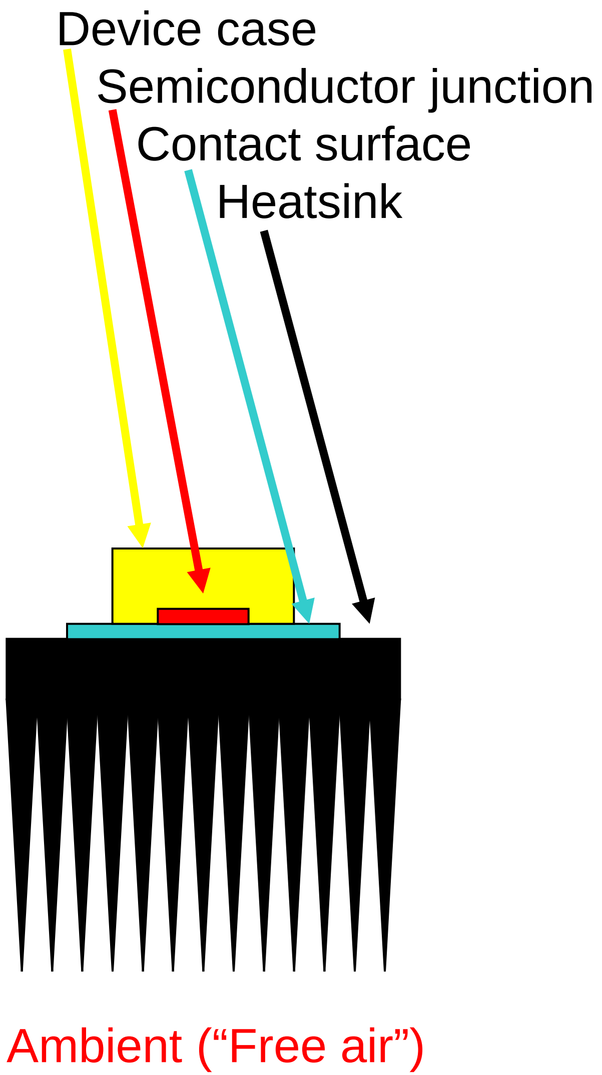
Thus, the IGBT combines:

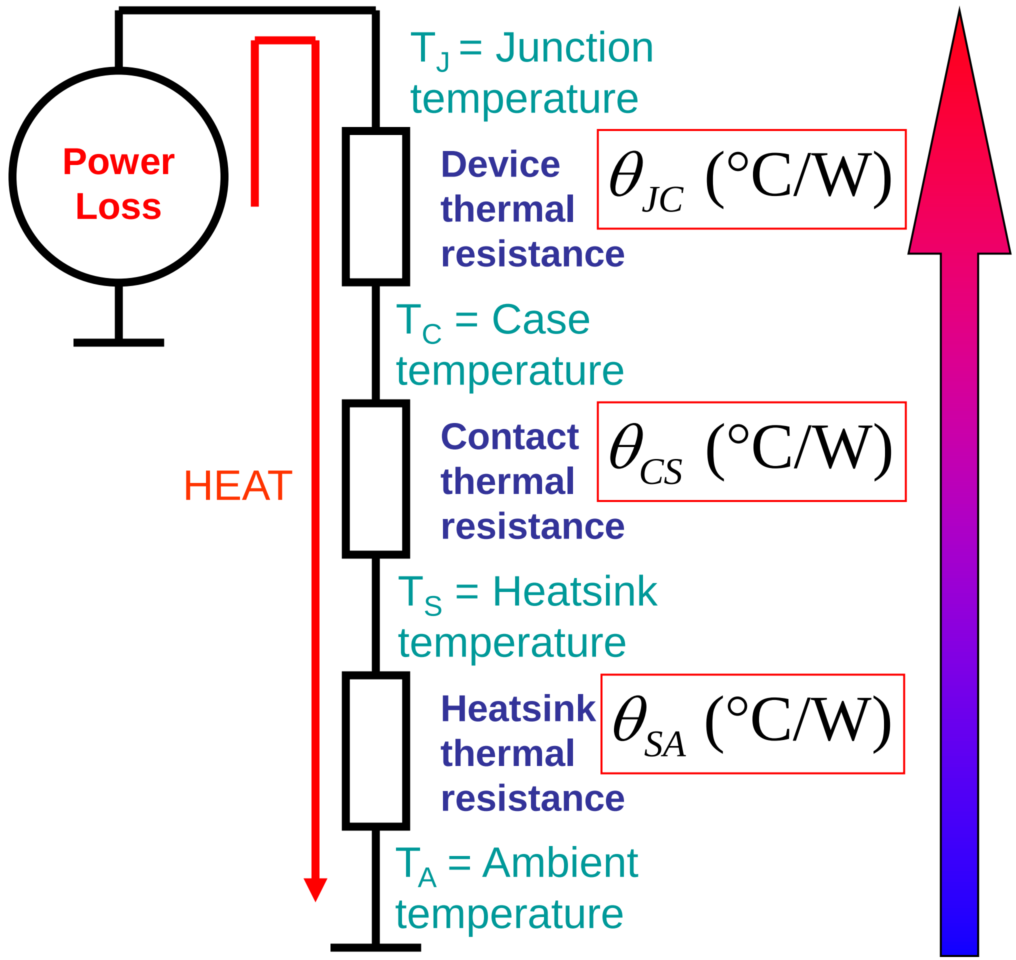
- MOSFET input characteristics (voltage-controlled, high input impedance)
- BJT conduction characteristics (low conduction loss at high voltage)

- **Two advantages of IGBTs over power metal-oxide-semiconductor field-effect transistors (MOSFETs) in medium- to high-power applications.**

1. Lower conduction loss at high voltage (>400 V) due to the conductivity modulation of the drift region.
2. Higher current handling capability. Suitable for medium and high power applications (e.g., motor drives, inverters).

Equivalent Thermal Model





$$\boxed{\Delta\,\mathrm{Temp}\,(°\mathrm{C}) = \mathrm{Power}\,(\mathrm{W}) \times \sum \theta_{\mathrm{XX}}}$$

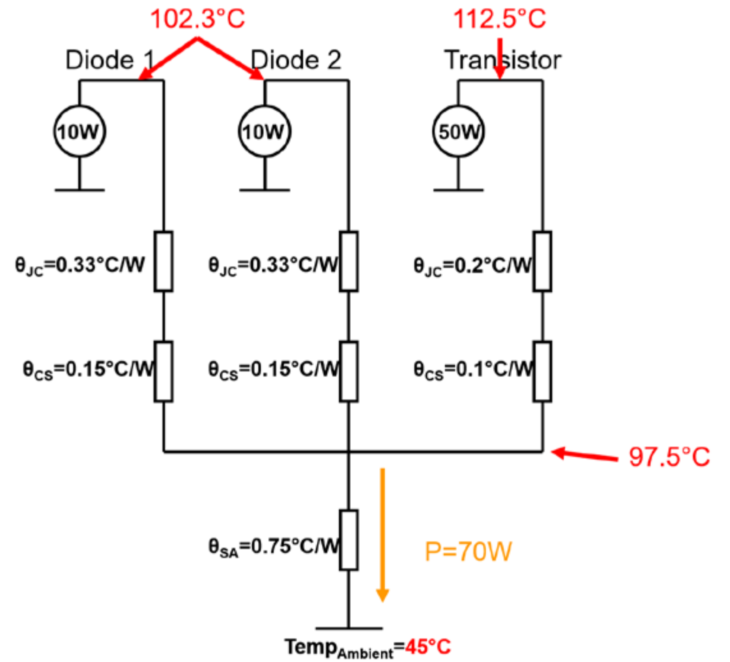
The heatsink needs to have low thermal resistance to the surrounding cooler air.

Example 1.

Two diodes and a transistor share a single heatsink. The thermal resistances are given in the table below. What is the junction temperature for each device if the ambient temperature is 45°C and the heatsink thermal resistance is 0.75°C/W?

Each Diode	Transistor
θ_{JC}=0.33°C/W	θ_{JC}=0.2°C/W
θ_{CS}=0.15°C/W	θ_{CS}=0.1°C/W
Power = 10W	Power = 50W

Thermal circuit diagram



First, calculate the heatsink temperature:

$T_{sink} = T_{amb} + P_{total} \times \theta_{SA} = 45 + 70W \times 0.75 = 97.5°C$

Now calculate the diode junction temperatures:

$T_{Diode} = T_{sink} + P_{diode} \times (\theta_{JC} + \theta_{CS})$

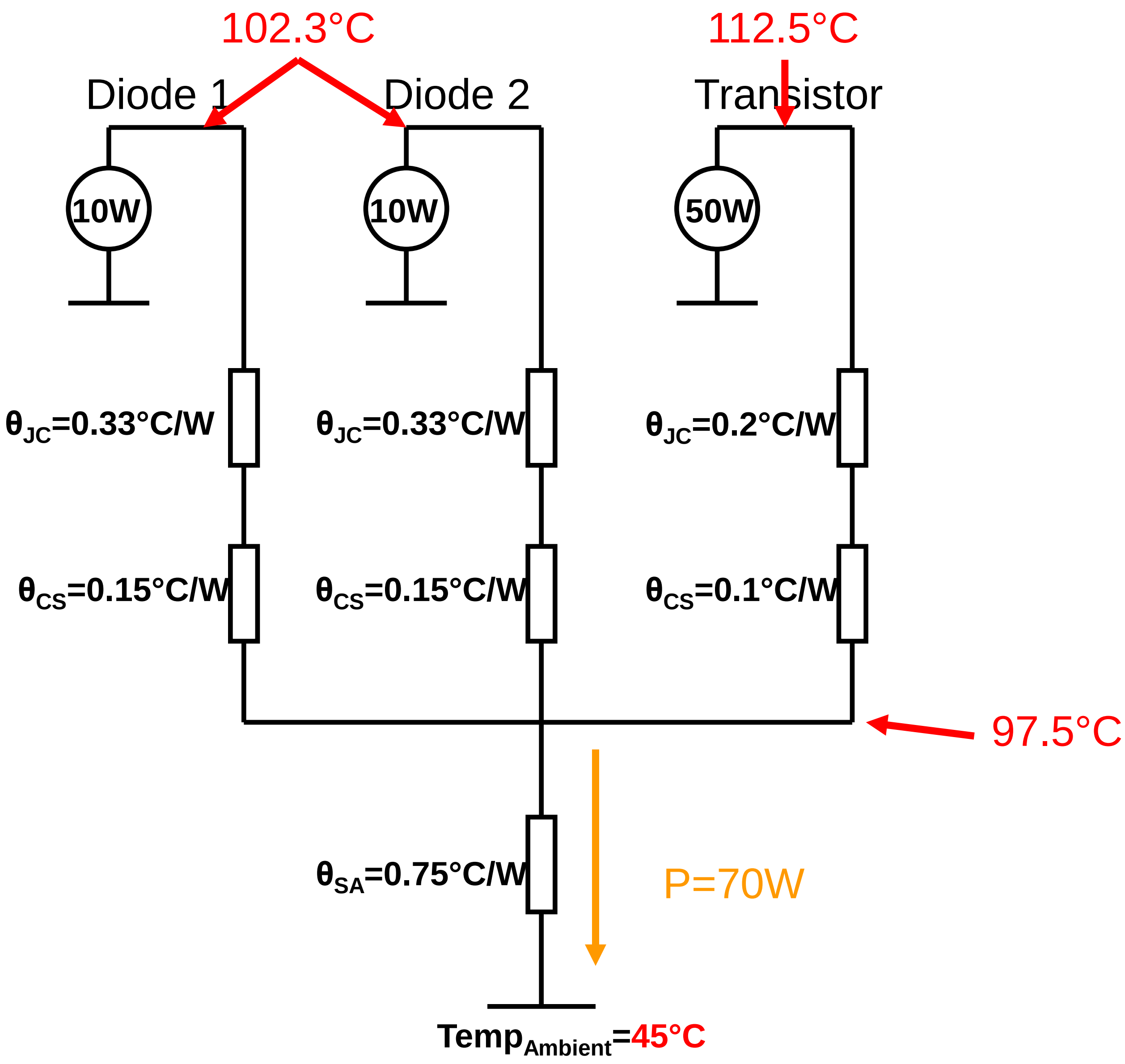
$= 97.5°C + 10 \times 0.48 = 102.3°C$

Finally, calculate the transistor junction temperature:

$T_{trans} = T_{sink} + P_{trans} \times (\theta_{JC} + \theta_{CS})$

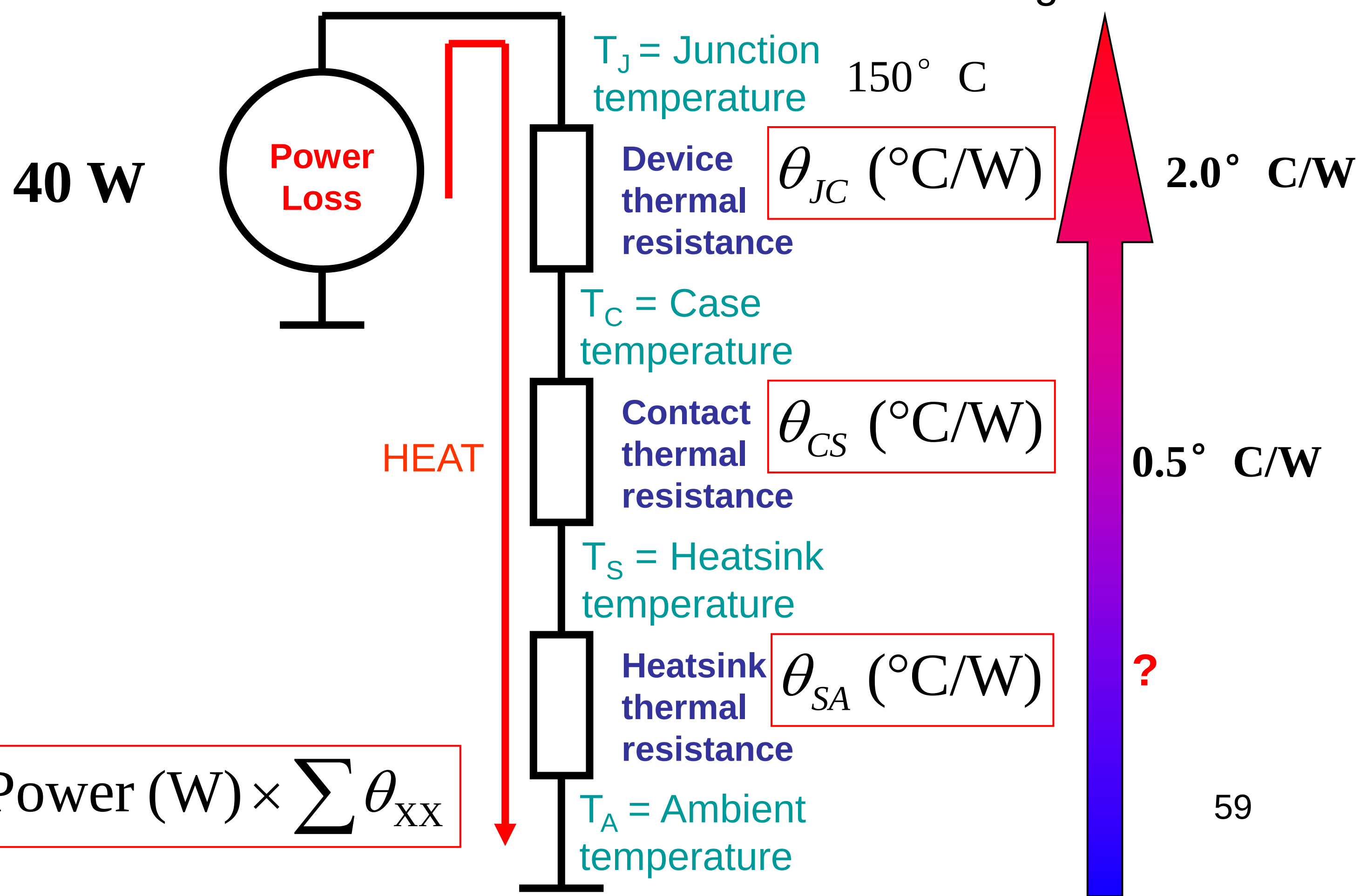
$= 97.5°C + 50 \times 0.3 = 112.5°C$

Thermal circuit diagram for example 1.



- To avoid a risk of overheating, a TIP120 transistor is required to dissipate 40 W in an ambient temperature of 30° C. Determine the thermal resistance of the heatsink required to keep the junction temperature below 150° C. There is an insulating washer of thermal resistance 0.5° C/W between the case and the heatsink. The transistor is specified for a dissipation of 65 W below 25° C, derated at 0.5 W/° C at higher temperatures. [Note that derating is the inverse of the thermal resistance and is an alternative specification.]

Thermal circuit diagram



$$\boxed{\Delta\,\text{Temp}\,(^\circ\text{C}) = \text{Power}\,(\text{W}) \times \sum \theta_{XX}}$$